Mac OS X File System Maintenance With

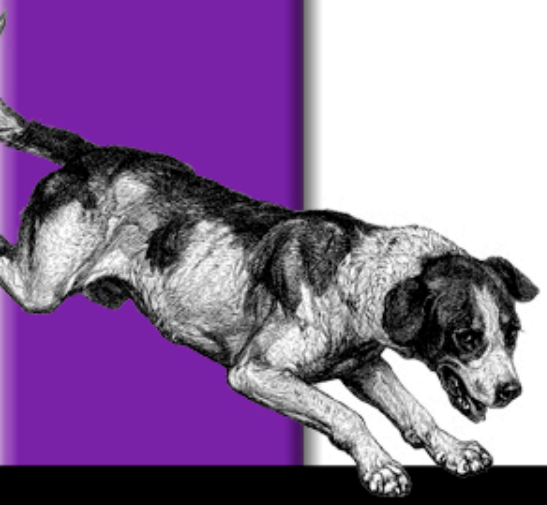
Radmind



By
Richard Glaser
University of Utah



O'Reilly Mac OS X Conference
October 25—28, 2004



O'REILLY®
Mac OS® X
Conference

Who am I?

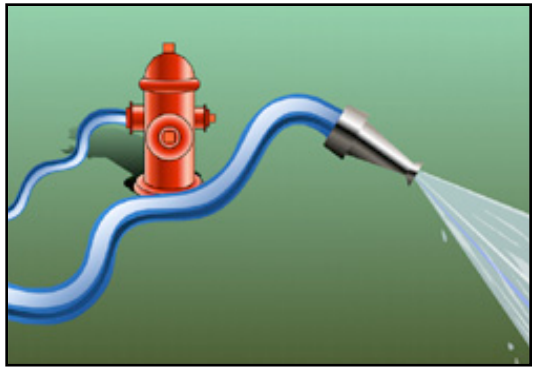
- System Administrator at University of Utah
  - Manage around 400 Mac OS X clients
  - Manage many servers that run multiple services
  - Provide students, staff & faculty support & training
  - Provide campus manager support & training
  - And much more...
- Long time Radmind Administrator
  - Have used it since May, 2002
- Steering Member of the MacEnterprise Project
  - Formerly MacOSXLabs Project



MacEnterprise.org
Mac OS X enterprise deployment project

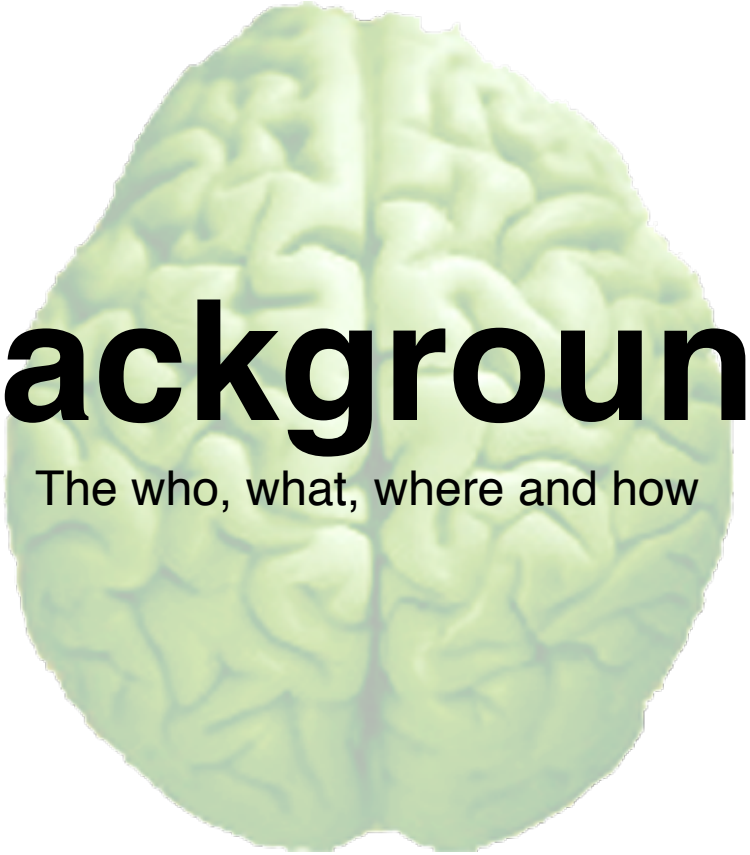
What I will cover

- Background
- The Basic Process
- Why use it?
- How to deploy?
- Terms
- Overview of command line tools
- Brief Tutorial





A lot of material to cover...
Sorry to make you drink from a fire hose



Background

The who, what, where and how



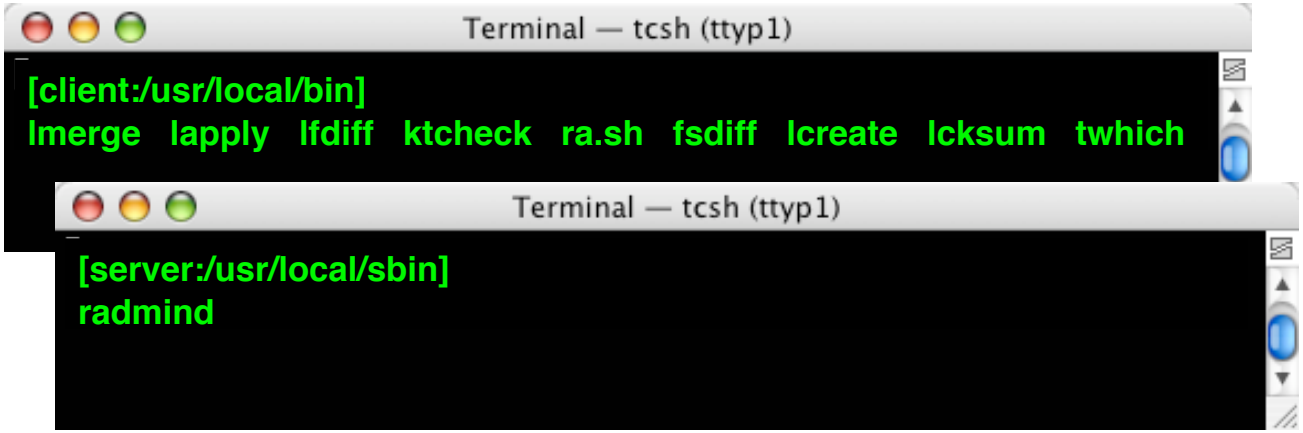
What is it?

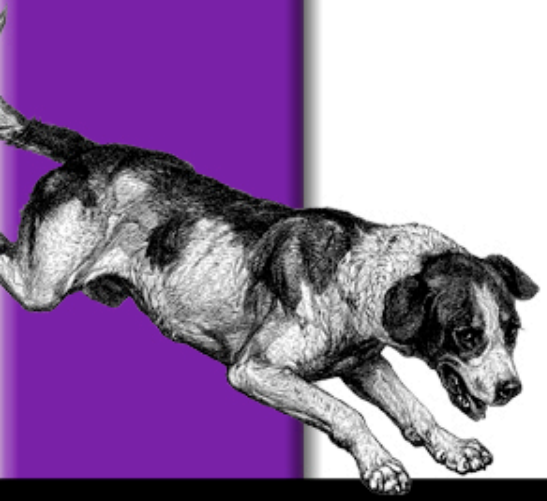
- Suite of command line tools
  - Both client and server





  - Remote Administration
    - Allows remote administration of multiple clients file system
  - Tripwire
    - At its core, it is a tripwire, detecting file system changes
  - Reverses Changes
    - But goes beyond just detecting changes it reverses them





Where to get it?

- Download from radmind web site:
  - www.radmind.org



- Mac OS X Installer



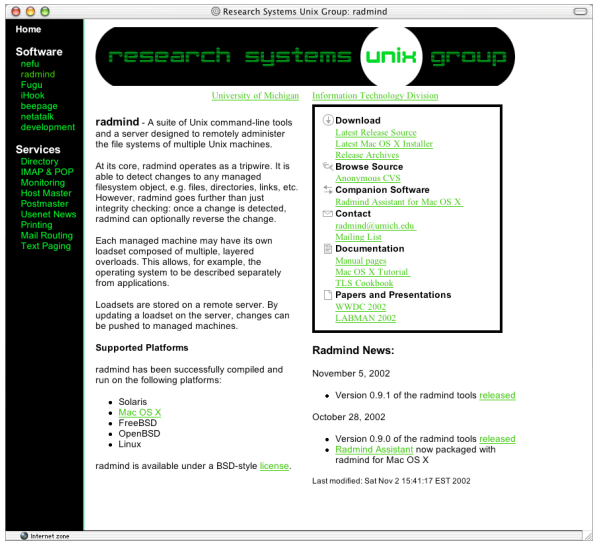
- Source Code



- Radmind Assistant
  (GUI front-end for radmind)



- iHook
  (A useful companion tool)





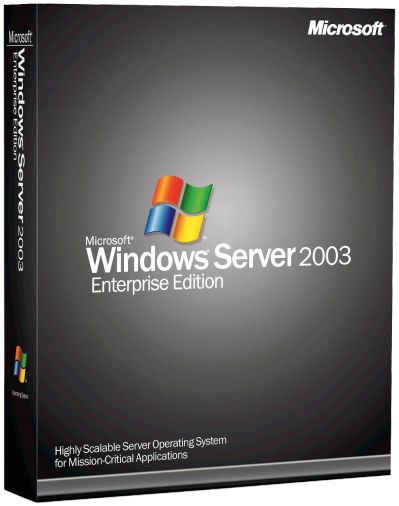
What platforms does it support?

- Can be used with any supported Unix
  - Mac OS X
  - FreeBSD
  - Linux
  - OpenBSD
  - Solaris



Other Platforms?

- Project to port radmind to Windows
  - Organized by:
    - University of Michigan & Stanford University
  - Others can join project to help code & test port
  - Mail list is setup called radmind-pc
    - To subscribe send email to majordomo@lists.stanford.edu and put in the body subscribe radmind-pc





How much is it?

- Open Source
- Available under BSD-style license



  - Can be used for **FREE** if:

    - You **ARE NOT** making profit from it and...
    - **Include** copyright & permision notice with distributed copies & documentation

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Permission to use, copy, modify, and distribute this software and its documentation for any purpose and without fee is hereby granted, provided that the above copyright notice appears in all copies and that both that copyright notice and this permission notice appear in supporting documentation, and that the name of The University of Michigan not be used in advertising or publicity pertaining to distribution of the software without specific, written prior permission. This software is supplied as is without expressed or implied warranties of any kind.





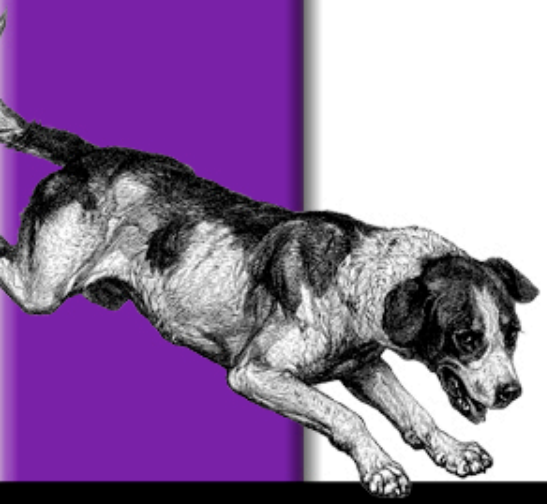
Who Developing It?

- Developed at the University of Michigan
  - By Research Systems Unix Group (RSUG)
    - Very responsive to bug reports & features requests
    - Active on mail list providing support & suggestions
  - Email

    radmind@umich.edu



  - research systems unix group
  - Its open source project, so anybody can contribute



Resources



- Mail list at Rice University:
  - radmind-discussion
    https://mailman.rice.edu/mailman/listinfo/radmind
  - radmind-announce
    https://mailman.rice.edu/mailman/listinfo/radmind-announce
  - Web form to subscribe to radmind lists



- Resources web page
  - At University of Utah:
    http://www.macos.utah.edu/Documentation/radmind/resources.html
    - Documentation
    - Mail Lists
    - Presentations
    - Third Party Tools
    - WebCasts
    - Web Sites



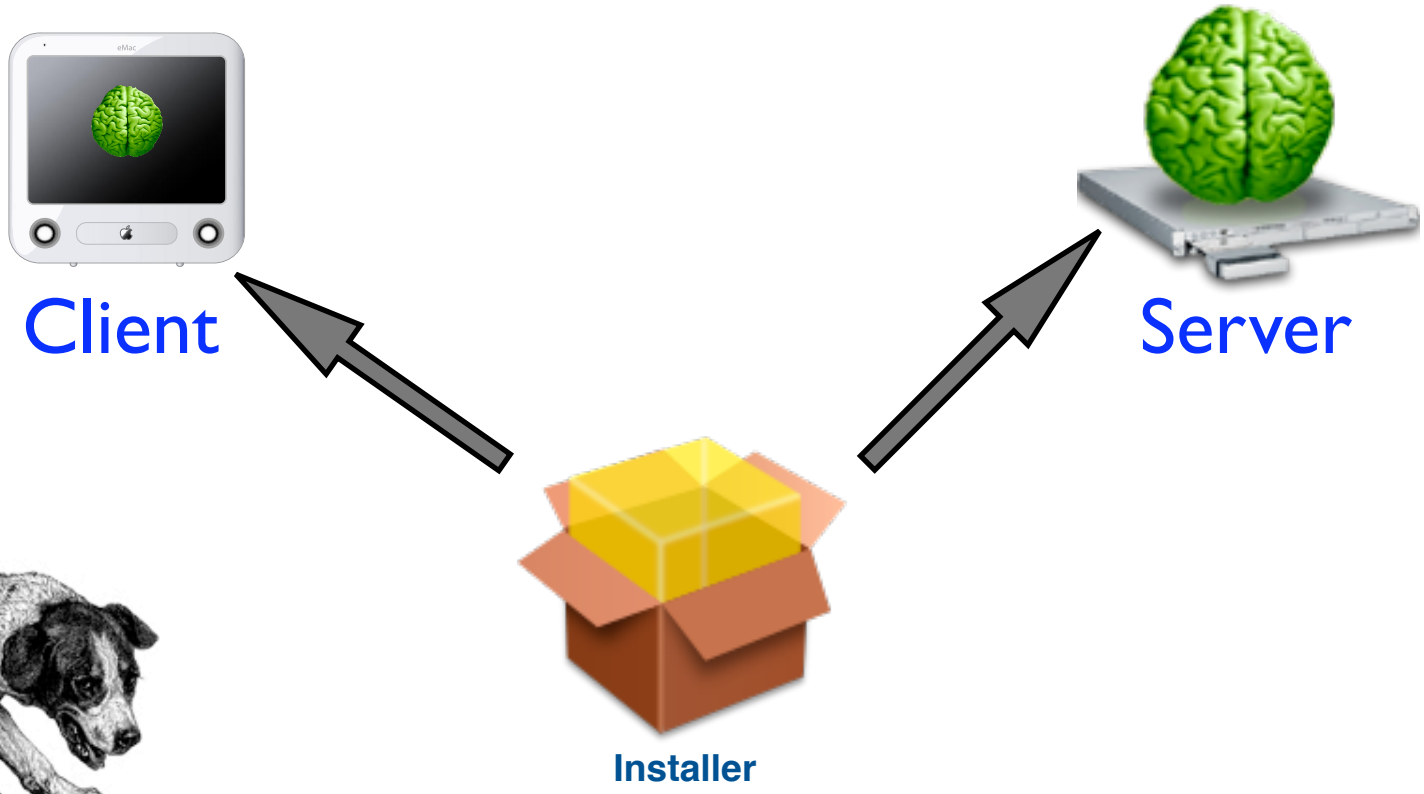
The Basic Process

The basics on how radmind works



Setup & Installation

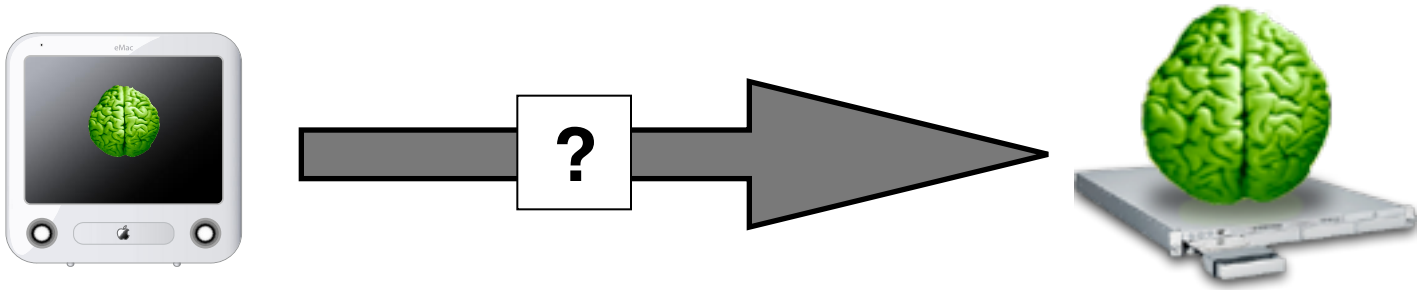
- Client & Server with radmind setup & installed
  - More to come on this later...





Client Requests File System

- Client requests managed file system from server





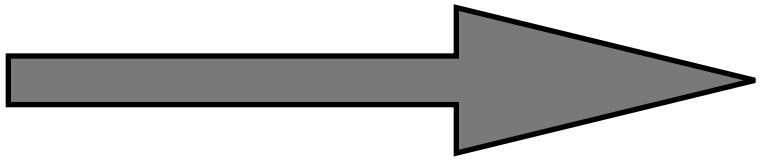
Is client managed?

- Server checks if client managed?









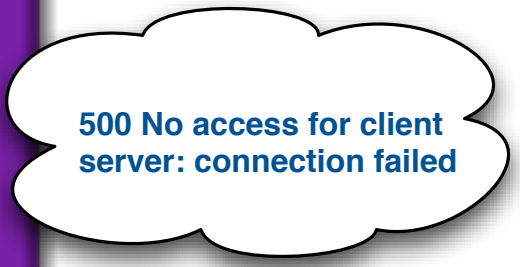


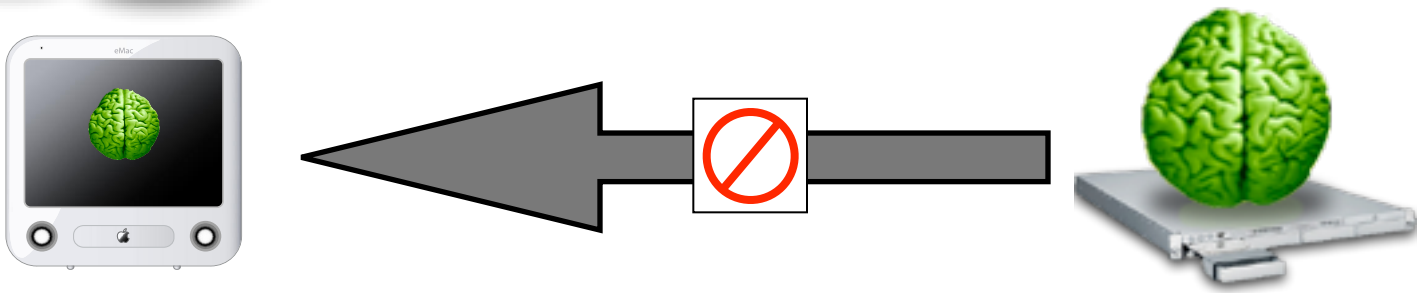
Client

Server

Client NOT Managed

- If client is NOT managed
  - Server rejects connection from client





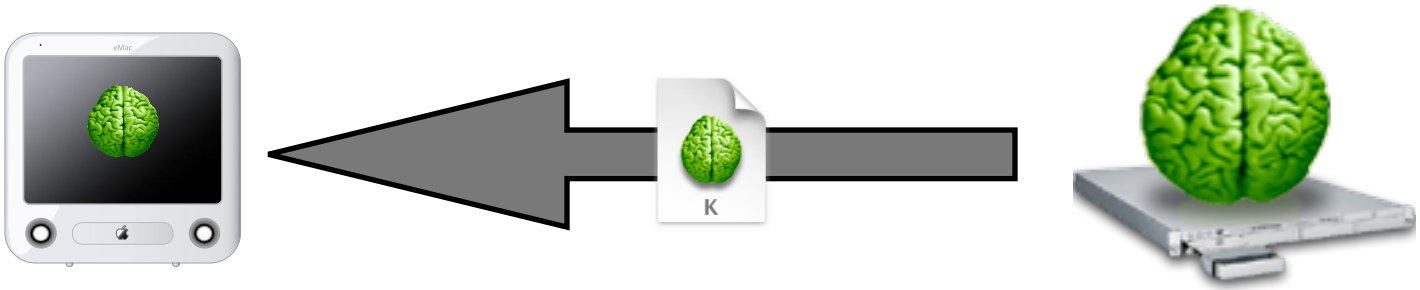
Client

Server



Client Managed - Sends Description

- If client is managed,
  - Server sends client managed file system description.





Client Stores Description

- Client stores managed file system description



Client



Server





/var/radmind/client

Client Compares Local & Managed

- Next, client scans local file system
- And compares against managed file system

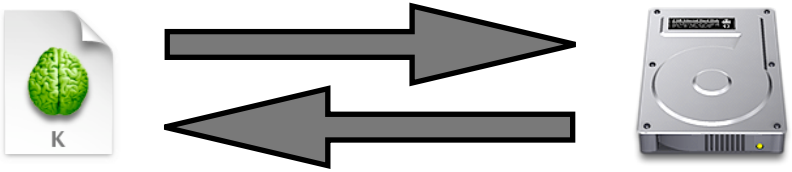


Client



Server





Managed File System Description

Client File System

Client Creates Differences

- Then creates a differences description



Client



Server



Differences Description



Managed File System Description



Client File System



Applies Differences to Client

- Using the differences description...
- It removes additional items on local file system



Client



Server



Differences Description

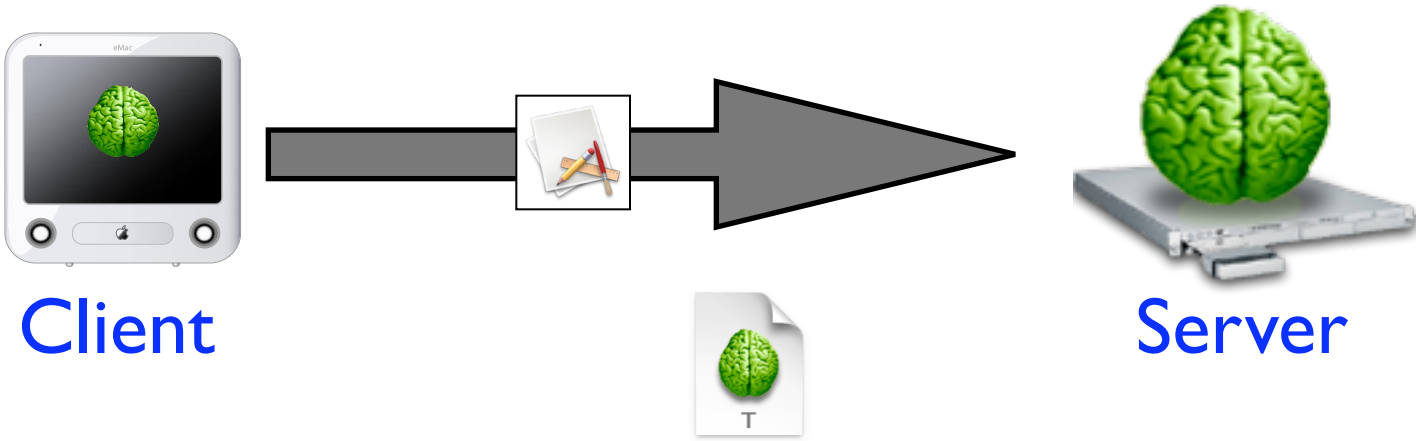




Removes Additional Items

Client Requests Missing Items

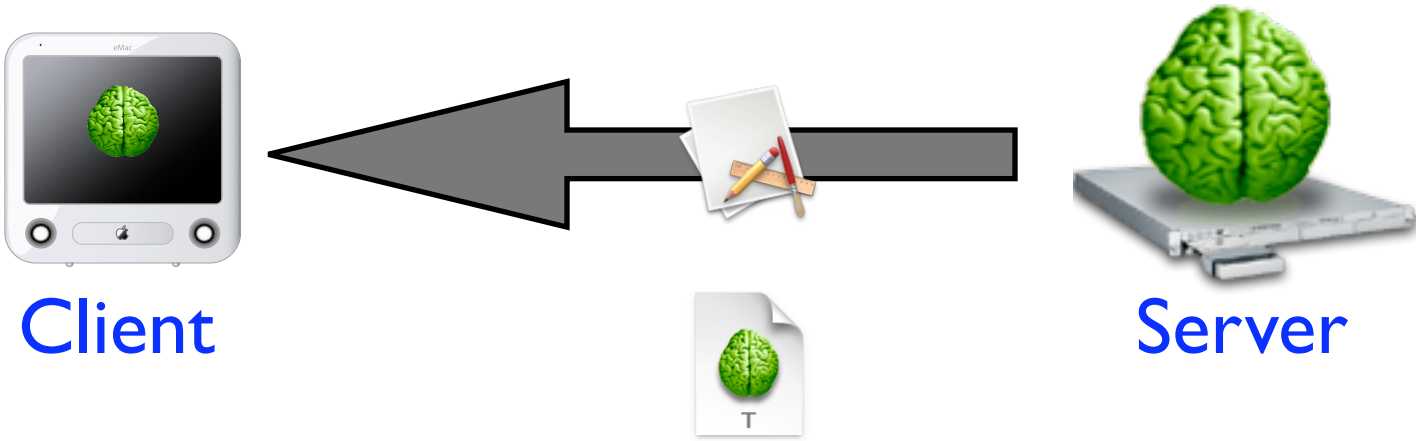
- If item(s) are missing, client request items from server





Server Sends Requested Items

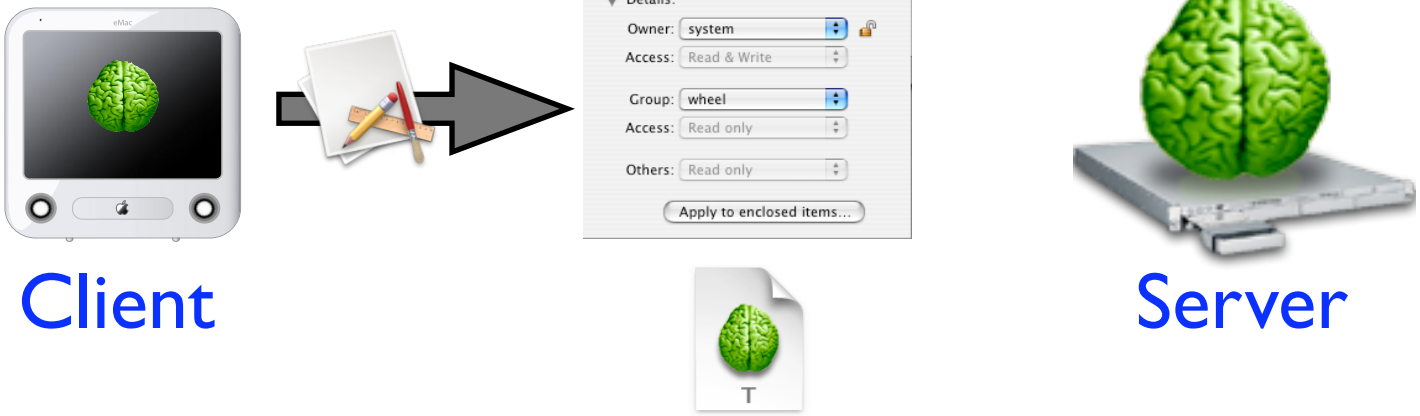
- And item(s) is downloaded from server





Updates Permissions & Attributes

- Then modifies permissions or attributes on items





Client Updated to Match Managed

- DONE!!!
  - Client file system updated to match managed

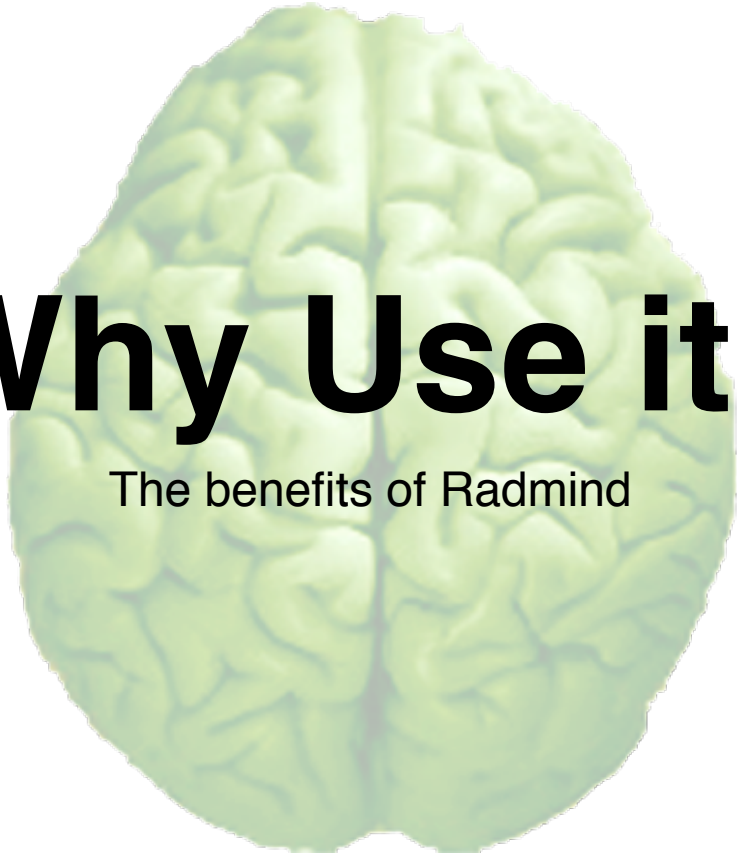


Client



Server





Why Use it?

The benefits of Radmind



Why use it? (Part I)



- Security
  - Quickly find & fix security issues
  - Tripwire & Reverse Changes
  - SSL Authentication & Encryption

OpenSSL

- Manage Nomadic Machines
  - Manage laptops & dynamically assigned Mac's
  - Identify clients using certificates



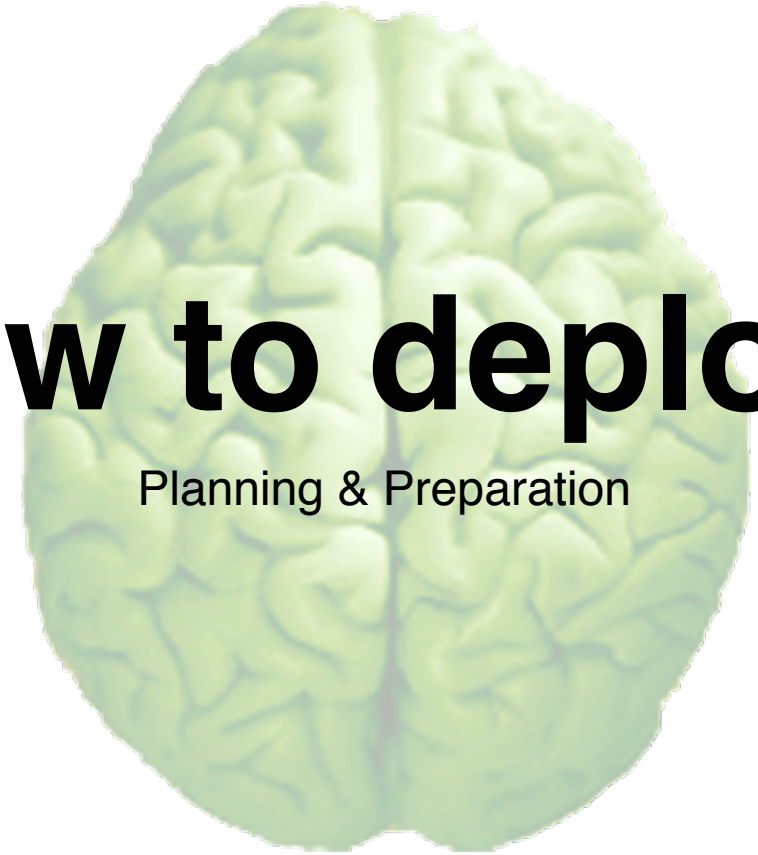
- Network & Server Utilization
  - Tools minimize network/server traffic
  - Few files transfered for incremental updates
  - Comparison done locally NOT over network



Why use it? (Part II)

- Package Management
  - Track & Test installation
  - Control distribution & installation
  - Fix, upgrade, downgrade or remove distribution
- Crappy Apps
  - Troubleshoot enterprise unfriendly applications
  - World Writable, SUID/GUID, Permissions, Attributes
  - Solve installation & distribution issues
- Granularity
  - Control & customized Application & OS distribution
  - Layered file system control
- Easy Management
  - One file on server to manage clients file system
  - Quickly change file systems between clients





How to deploy?

Planning & Preparation



How to deploy it? (Part I)

- Environment
  - Student, staff or server environments
- Policy
  - Administrative policy & procedures
  - What you want to allow user to do or not do?
  - Allow staff be local admin?
- Updating Client
  - How often?
  - Automated vs Manual?
  - Allow users to manually initiate updates?
- File System Management
  - Broad vs Granular
  - Less Complex vs Flexibility, Power & Customization



How to deploy it? (Part II)

- Client File System Organization
  - Consider & define naming scheme
  - Building or Lab? Purpose? Person's Name?
- File System Descriptions Organization
  - Application vs OS vs administrative vs customization
- Separate Software System Components
  - Reduce dependency on software with system
  - Usually Apple software
  - Allows easier migration to new OS's
- Application Dependency
  - Some software shares/modifies same items
  - Ex - Adobe Registration Database





Terms

Radmind Lingo & Slang



Transcript

- A plain text file that describes file system objects
- Each line describes a objects like:
  - Files, directories, links, etc.



- A transcript can describe...
  - The entire file system
  - Installs or updates
    - Like Office 2004 or Security Update 2004-30-20
  - Individual files, directories, links, etc.
    - /Application
    - ~/Library/Preferences/com.apple.dock.plist



- Transcript name format
  - <transcript_name>.T (app_office_2004.T)
  - Can't use spaces in name



Transcript

- A transcript includes the following items:
  - type
  - path
  - mode
  - uid
  - gid
  - mtime
  - size
  - checksum



Transcript

- A transcript has the following format:

  type path mode uid gid mtime size checksum

- Type is a single letter describing file system objects
  - **a** applefile (Mac OS multi-forked files)
  - **b** block special file
  - **c** character special file
  - **d** directory
  - **D** door (Solaris)
  - **f** file
  - **h** hard link
  - **l** symbolic link
  - **p** pipe
  - **s** socket



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Transcript

- A transcript has the following format:

  type **path** mode uid gid mtime size checksum

- Path is a encoded path:
  - **\b** space
  - **\t** tab
  - **\n** newline
  - **\r** carriage return
  - **** \\



Transcript

- A transcript has the following format:

  type **path** mode uid gid mtime size checksum

- Path is a encoded path:
  - **\b** space (/Applications/Chess\b2.0.app)
  - **\t** tab
  - **\n** newline
  - **\r** carriage return
  - **** \\



Transcript

- A transcript has the following format:

  type path **mode** uid gid mtime size checksum

- Mode is the octal representation of the permissions

- Examples:

  - 0775
  - 0644



Transcript

- A transcript has the following format:

  type path mode **uid** gid mtime size checksum

- User ID

- Examples:

  – 0



Transcript

- A transcript has the following format:

  type path mode uid **gid** mtime size checksum

- Group ID
- Examples:
  - 80



Transcript

- A transcript has the following format:

  type path mode uid gid **mtime** size checksum

- Modification time
  - mtime is the number of seconds since 1970 GMT
  - Analogous to modification date

- Examples:

  - 1063587946
  - 1063586386



Transcript

- A transcript has the following format:

  type path mode uid gid mtime **size** checksum

- Size
  - Size of the file in bytes
- Examples:
  - 13207
  - 8254
  - 29009



Transcript

- A transcript has the following format:

  type path mode uid gid mtime size **checksum**

- A value based on the contents of data
- Is the base64 encoded if enabled, otherwise it is "-".

- sha1 encoding is default
  - Others can be used:
    - sha
    - md5
    - md2
    - dss1
    - mdc2
    - ripemd160



Transcript

- Example transcript for Chess

type **path** **mode** uid gid mtime size checksum

```
d /Applications/Chess\b2.0.app                                0775 501 80
d /Applications/Chess\b2.0.app/Contents                       0775 501 80
a /Applications/Chess\b2.0.app/Contents/.DS_Store             0664 501 80 1069114356 6242   Z6YoEpVu0s2ld31tpKchl7JntKk=
a /Applications/Chess\b2.0.app/Contents/Info.plist            0664 501 80 1063446253 1829   a8NJHudyyx6pfT9oHNVaykNSiNc=
d /Applications/Chess\b2.0.app/Contents/MacOS                 0775 501 80
a /Applications/Chess\b2.0.app/Contents/MacOS/Chess           0775 501 80 1096299220 220822 g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a /Applications/Chess\b2.0.app/Contents/PkgInfo               0664 501 80 1063446253 102    9Kw3S/TT9JC9BoRJ5HbGF9ld2vw=
d /Applications/Chess\b2.0.app/Contents/Resources             0775 501 80
a /Applications/Chess\b2.0.app/Contents/Resources/1.rgb       0664 501 80 1063446253 2740   K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a /Applications/Chess\b2.0.app/Contents/Resources/2.rgb       0664 501 80 1063446253 3037   ZocJtCPnu0NH0BSw92MaqlBFeSA=
a /Applications/Chess\b2.0.app/Contents/Resources/3.rgb       0664 501 80 1063446253 3040   CGZPshzPPOoQyfbNZqnWUTal7W4=
a /Applications/Chess\b2.0.app/Contents/Resources/4.rgb       0664 501 80 1063446253 2971   KlkvasubxAFrT4RAsnHN7TR2bQQ=
a /Applications/Chess\b2.0.app/Contents/Resources/5.rgb       0664 501 80 1063446253 3082   w/C/hB03jMAVVG8KYXR3RzyNcns=
a /Applications/Chess\b2.0.app/Contents/Resources/6.rgb       0664 501 80 1063446253 3157   iklyWKAmpJHfbiF5uWFY3djkfB4=
a /Applications/Chess\b2.0.app/Contents/Resources/7.rgb       0664 501 80 1063446253 2959   6/vX4bgrKktSHlvSE4eloMc1lCk=
a /Applications/Chess\b2.0.app/Contents/Resources/8.rgb       0664 501 80 1063446253 3052   zmRPzkAgoQHnZmCxXhkASFvKNVI=
a /Applications/Chess\b2.0.app/Contents/Resources/COPYING     0664 501 80 1063446253 18087  UlEtM4SgllBznd0YzZZiy/3RYRk=
```



Transcript

- Example transcript for Chess

type **path** **mode** uid gid mtime size checksum

```
d ./Applications/Chess\b2.0.app                                  0775  501  80
d ./Applications/Chess\b2.0.app/Contents                         0775  501  80
a ./Applications/Chess\b2.0.app/Contents/.DS_Store               0664  501  80  1069114356  6242    Z6YoEpVu0s2ld31tpKchl7JntKk=
a ./Applications/Chess\b2.0.app/Contents/Info.plist              0664  501  80  1063446253  1829    a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS                   0775  501  80
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess             0775  501  80  1096299220  220822  g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo                 0664  501  80  1063446253  102     9Kw3S/TT9JC9BoRJ5HbGF9ld2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources               0775  501  80
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb         0664  501  80  1063446253  2740    K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb         0664  501  80  1063446253  3037    ZocJtCPnu0NH0BSw92MaqlBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb         0664  501  80  1063446253  3040    CGZPshzPPOoQyfbNZqnWUTal7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb         0664  501  80  1063446253  2971    KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb         0664  501  80  1063446253  3082    w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb         0664  501  80  1063446253  3157    iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb         0664  501  80  1063446253  2959    6/vX4bgrKktSHlvSE4eloMc1lCk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb         0664  501  80  1063446253  3052    zmRPzkAgoQHnZmCxXhkASFvKNVI=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING       0664  501  80  1063446253  18087   UlEtM4SgllBznd0YzZZiy/3RYRk=
```



Transcript

- Example transcript for Chess

type path **mode** uid gid mtime size checksum

d ./Applications/Chess\b2.0.app	0775	501	80			
d ./Applications/Chess\b2.0.app/Contents	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/.DS_Store	0664	501	80	1069114356	6242	Z6YoEpVu0s2Id31tpKchI7JntKk=
a ./Applications/Chess\b2.0.app/Contents/Info.plist	0664	501	80	1063446253	1829	a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess	0775	501	80	1096299220	220822	g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo	0664	501	80	1063446253	102	9Kw3S/TT9JC9BoRJ5HbGF9Id2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb	0664	501	80	1063446253	2740	K2+Ub1bA9iArGBkI7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb	0664	501	80	1063446253	3037	ZocJtCPnu0NH0BSw92MaqIBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb	0664	501	80	1063446253	3040	CGZPshzPPOoQyfbNZqnWUTaI7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb	0664	501	80	1063446253	2971	KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb	0664	501	80	1063446253	3082	w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb	0664	501	80	1063446253	3157	iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb	0664	501	80	1063446253	2959	6/vX4bgrKktSHlvSE4eloMc1lCk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb	0664	501	80	1063446253	3052	zmRPzkAgoQHnZmCxXhkASFvKNVI=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING	0664	501	80	1063446253	18087	UlEtM4SgIlBznd0YzZZiy/3RYRk=



Transcript

- Example transcript for Chess

type path mode **uid** gid mtime size checksum

```
d ./Applications/Chess\b2.0.app                                0775  501  80
d ./Applications/Chess\b2.0.app/Contents                       0775  501  80
a ./Applications/Chess\b2.0.app/Contents/.DS_Store             0664  501  80  1069114356  6242    Z6YoEpVu0s2ld31tpKchl7JntKk=
a ./Applications/Chess\b2.0.app/Contents/Info.plist            0664  501  80  1063446253  1829    a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS                 0775  501  80
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess           0775  501  80  1096299220  220822  g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo               0664  501  80  1063446253  102     9Kw3S/TT9JC9BoRJ5HbGF9ld2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources             0775  501  80
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb       0664  501  80  1063446253  2740    K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb       0664  501  80  1063446253  3037    ZocJtCPnu0NH0BSw92MaqlBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb       0664  501  80  1063446253  3040    CGZPshzPPOoQyfbNZqnWUTal7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb       0664  501  80  1063446253  2971    KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb       0664  501  80  1063446253  3082    w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb       0664  501  80  1063446253  3157    iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb       0664  501  80  1063446253  2959    6/vX4bgrKktSHlvSE4eloMc1lCk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb       0664  501  80  1063446253  3052    zmRPzkAgoQHnZmCxXhkASFvKNVI=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING     0664  501  80  1063446253  18087   UlEtM4SgIlBznd0YzZZiy/3RYRk=
```



Transcript

- Example transcript for Chess

type path mode uid **gid** mtime size checksum

```
d ./Applications/Chess\b2.0.app                                0775  501  80
d ./Applications/Chess\b2.0.app/Contents                       0775  501  80
a ./Applications/Chess\b2.0.app/Contents/.DS_Store             0664  501  80  1069114356  6242    Z6YoEpVu0s2ld31tpKchl7JntKk=
a ./Applications/Chess\b2.0.app/Contents/Info.plist            0664  501  80  1063446253  1829    a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS                 0775  501  80
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess           0775  501  80  1096299220  220822  g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo               0664  501  80  1063446253  102     9Kw3S/TT9JC9BoRJ5HbGF9ld2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources             0775  501  80
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb       0664  501  80  1063446253  2740    K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb       0664  501  80  1063446253  3037    ZocJtCPnu0NH0BSw92MaqlBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb       0664  501  80  1063446253  3040    CGZPshzPPOoQyfbNZqnWUTal7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb       0664  501  80  1063446253  2971    KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb       0664  501  80  1063446253  3082    w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb       0664  501  80  1063446253  3157    iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb       0664  501  80  1063446253  2959    6/vX4bgrKktSHlvSE4eloMc1lCk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb       0664  501  80  1063446253  3052    zmRPzkAgoQHnZmCxXhkASFvKNVI=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING     0664  501  80  1063446253  18087   UlEtM4SgIlBznd0YzZZiy/3RYRk=
```



Transcript

- Example transcript for Chess

type path mode uid gid **mtime** size checksum

d	./Applications/Chess\b2.0.app	0775	501	80			
d	./Applications/Chess\b2.0.app/Contents	0775	501	80			
a	./Applications/Chess\b2.0.app/Contents/.DS_Store	0664	501	80	1069114356	6242	Z6YoEpVu0s2ld31tpKchl7JntKk=
a	./Applications/Chess\b2.0.app/Contents/Info.plist	0664	501	80	1063446253	1829	a8NJHudyyx6pfT9oHNVaykNSiNc=
d	./Applications/Chess\b2.0.app/Contents/MacOS	0775	501	80			
a	./Applications/Chess\b2.0.app/Contents/MacOS/Chess	0775	501	80	1096299220	220822	g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a	./Applications/Chess\b2.0.app/Contents/PkgInfo	0664	501	80	1063446253	102	9Kw3S/TT9JC9BoRJ5HbGF9ld2vw=
d	./Applications/Chess\b2.0.app/Contents/Resources	0775	501	80			
a	./Applications/Chess\b2.0.app/Contents/Resources/1.rgb	0664	501	80	1063446253	2740	K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a	./Applications/Chess\b2.0.app/Contents/Resources/2.rgb	0664	501	80	1063446253	3037	ZocJtCPnu0NH0BSw92MaqlBFeSA=
a	./Applications/Chess\b2.0.app/Contents/Resources/3.rgb	0664	501	80	1063446253	3040	CGZPshzPPOoQyfbNZqnWUTal7W4=
a	./Applications/Chess\b2.0.app/Contents/Resources/4.rgb	0664	501	80	1063446253	2971	KlkvasubxAFrT4RAsnHN7TR2bQQ=
a	./Applications/Chess\b2.0.app/Contents/Resources/5.rgb	0664	501	80	1063446253	3082	w/C/hB03jMAVVG8KYXR3RzyNcns=
a	./Applications/Chess\b2.0.app/Contents/Resources/6.rgb	0664	501	80	1063446253	3157	iklyWKAmpJHfbiF5uWFY3djkfB4=
a	./Applications/Chess\b2.0.app/Contents/Resources/7.rgb	0664	501	80	1063446253	2959	6/vX4bgrKktSHlvSE4eloMc1lCk=
a	./Applications/Chess\b2.0.app/Contents/Resources/8.rgb	0664	501	80	1063446253	3052	zmRPzkAgoQHnZmCxXhkASFvKNVI=
a	./Applications/Chess\b2.0.app/Contents/Resources/COPYING	0664	501	80	1063446253	18087	UlEtM4SgllBznd0YzZZiy/3RYRk=



Transcript

- Example transcript for Chess

type path mode uid gid mtime **size** checksum

```
d ./Applications/Chess\b2.0.app                                0775  501  80
d ./Applications/Chess\b2.0.app/Contents                       0775  501  80
a ./Applications/Chess\b2.0.app/Contents/.DS_Store             0664  501  80  1069114356  6242    Z6YoEpVu0s2ld31tpKchl7JntKk=
a ./Applications/Chess\b2.0.app/Contents/Info.plist            0664  501  80  1063446253  1829    a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS                 0775  501  80
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess           0775  501  80  1096299220  220822  g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo               0664  501  80  1063446253  102     9Kw3S/TT9JC9BoRJ5HbGF9ld2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources             0775  501  80
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb       0664  501  80  1063446253  2740    K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb       0664  501  80  1063446253  3037    ZocJtCPnu0NH0BSw92MaqlBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb       0664  501  80  1063446253  3040    CGZPshzPPOoQyfbNZqnWUTal7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb       0664  501  80  1063446253  2971    KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb       0664  501  80  1063446253  3082    w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb       0664  501  80  1063446253  3157    iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb       0664  501  80  1063446253  2959    6/vX4bgrKktSHlvSE4eloMc1lCk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb       0664  501  80  1063446253  3052    zmRPzkAgoQHnZmCxXhkASFvKNVI=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING     0664  501  80  1063446253  18087   UlEtM4SgllBznd0YzZZiy/3RYRk=
```



Transcript

- Example transcript for Chess

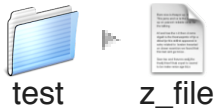
type path mode uid gid mtime size **checksum**

d ./Applications/Chess\b2.0.app	0775	501	80			
d ./Applications/Chess\b2.0.app/Contents	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/.DS_Store	0664	501	80	1069114356	6242	Z6YoEpVu0s2ld31tpKchl7JntKk=
a ./Applications/Chess\b2.0.app/Contents/Info.plist	0664	501	80	1063446253	1829	a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess	0775	501	80	1096299220	220822	g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo	0664	501	80	1063446253	102	9Kw3S/TT9JC9BoRJ5HbGF9ld2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb	0664	501	80	1063446253	2740	K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb	0664	501	80	1063446253	3037	ZocJtCPnu0NH0BSw92MaqlBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb	0664	501	80	1063446253	3040	CGZPshzPPOoQyfbNZqnWUTal7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb	0664	501	80	1063446253	2971	KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb	0664	501	80	1063446253	3082	w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb	0664	501	80	1063446253	3157	iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb	0664	501	80	1063446253	2959	6/vX4bgrKktSHlvSE4eloMc1lCk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb	0664	501	80	1063446253	3052	zmRPzkAgoQHnZmCxXhkASFvKNVI=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING	0664	501	80	1063446253	18087	UlEtM4SgllBznd0YzZZiy/3RYRk=



Transcripts - Sorting

- Transcripts are sorted alphabetically
  - Depth first, and case sensitively
  - This means subdirectories have precedence over files in the same directory.
  - For example, the file /test/z_file



  - Would come before: /test.old



  - Even though "." normally comes before "/"

```
f ./test/z_file      0644  501   20    1098059210  0    2jmj7l5rSw0yVb/vlWAYkK/YBwk=
f ./test.old         0644  501   20    1098059229  0    2jmj7l5rSw0yVb/vlWAYkK/YBwk=
```



Transcripts - Sorting

- Uppercase take precedence over lowercase
  - For example, the folder /Z_folder



  - Would come before: /a_folder



  - And both of these folders would come before, previous example file: /test.old

```
d ./Z_folder      0755  501   20
d ./a_folder      0755  501   20
f ./test/z_file   0644  501   20    1098059210  0    2jmj7l5rSw0yVb/vlWAYkK/YBwk=
f ./test.old      0644  501   20    1098059229  0    2jmj7l5rSw0yVb/vlWAYkK/YBwk=
```

  - You can use tools to fix sort order:
    - Command line tools like sort
    - GUI tools like BBEdit



Types of Transcripts

- Positive Transcript
  - Describes managed items & all attributes
- Negative Transcript
  - Describes unmanaged items with some attributes
- Special Transcript
  - Describes host specific items, like licenses files
  - Usually used with files that are tied to client hardware
- Create-able Transcript
  - Describes additions on client, like software installations
  - Upload to server & then distributed to other clients
- Apply-able Transcript
  - Describes differences on client, like end-user mods
  - Used to bring client to managed state



Positive Transcript

- Lists file system objects that **ARE** managed
- Radmind uses this transcript to check ALL attributes
  - Permissions, owner, group, mod time, file size
  - Optionally checksums are checked
- Contents of directories **ARE** read
- A positive transcript can describe items:
  - OS (Mac OS 10.3.5 or Mac OS X 10.2.8)
  - Software (Office 2004)
  - Software/OS (Security Update 2004-09-30)
  - Settings (Lab Printer configuration)
  - Individual Objects (Custom Dock - com.apple.dock.plist)



Positive Transcript

- Example of software “Chess 2.0”

d ./Applications/Chess\b2.0.app	0775	501	80			
d ./Applications/Chess\b2.0.app/Contents	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/.DS_Store	0664	501	80	1069114356	6242	Z6YoEpVu0s2Id31tpKchI7JntKk=
a ./Applications/Chess\b2.0.app/Contents/Info.plist	0664	501	80	1063446253	1829	a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess	0775	501	80	1096299220	220822	g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo	0664	501	80	1063446253	102	9Kw3S/TT9JC9BoRJ5HbGF9Id2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources	0775	501	80			
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb	0664	501	80	1063446253	2740	K2+Ub1bA9iArGBkI7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb	0664	501	80	1063446253	3037	ZocJtCPnu0NH0BSw92MaqIBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb	0664	501	80	1063446253	3040	CGZPshzPPOoQyfbNZqnWUTaI7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb	0664	501	80	1063446253	2971	KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb	0664	501	80	1063446253	3082	w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb	0664	501	80	1063446253	3157	iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb	0664	501	80	1063446253	2959	6/vX4bgrKktSHlvSE4eloMc1ICk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb	0664	501	80	1063446253	3052	zmRPzkAgoQHnZmCxXhkASFvKNVl=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING	0664	501	80	1063446253	18087	UlEtM4SgllBznd0YzZZiy/3RYRk=



Positive Transcript

- Example of OS update “Security Update 2004-09-30”

```
f ./System/Library/CoreServices/AppleFileServer.app/Contents/Info.plist 0644 0 0 1096354193 805
f ./System/Library/CoreServices/AppleFileServer.app/Contents/MacOS/AppleFileServer 0755 0 0 1097080184 1460964
f ./System/Library/CoreServices/AppleFileServer.app/Contents/PkgInfo 0644 0 0 1096354193 8
f ./System/Library/CoreServices/AppleFileServer.app/Contents/version.plist 0644 0 0 1096415934 461
d ./System/Library/QuickTime/QuickTimeBMPUpdate.component 0755 0 0
d ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents 0755 0 0
f ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/Info.plist 0644 0 0 1096414340 955
d ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/MacOS 0755 0 0
f ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/MacOS/QuickTimeBMPUpdate 0755 0 0 1097080184 18928
f ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/PkgInfo 0644 0 0 1096414340 8
d ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/Resources 0755 0 0
d ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/Resources/English.lproj 0755 0 0
f ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/Resources/English.lproj/InfoPlist.strings 0644 0 0 1096414340 96
f ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/Resources/English.lproj/Localized.rsrc 0644 0 0 1096414340 368
f ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/Resources/QuickTimeBMPUpdate.rsrc 0644 0 0 1096414340 732
f ./System/Library/QuickTime/QuickTimeBMPUpdate.component/Contents/version.plist 0644 0 0 1096416853 463
f ./Users/mac/Applications/Apple/NetInfo\bManager.app/Contents/Info.plist 0664 0 80 1095986128 1301
f ./Users/mac/Applications/Apple/NetInfo\bManager.app/Contents/MacOS/NetInfo\bManager 0775 0 80 1096415977 337640
f ./Users/mac/Applications/Apple/NetInfo\bManager.app/Contents/version.plist 0664 0 80 1096415977 466
f ./usr/libexec/postfix/smtpd 0755 0 0 1097080184 289808
f ./usr/sbin/cupsd 0755 0 0 1097080184 219864
```



Positive Transcript

- Example of settings (Lab Printer Config)

```
f ./Library/Preferences/.GlobalPreferences.plist  0644  0    80   1079131516  9319   Nzi5tGoiQDkNJ+ibmpEuT2GSe/4=
f ./Users/mac/.lpoptions                          0644  501  20   1079131517  23     1rwlsnjWK1+yjbb154i3fMeQXOc=
f ./Users/mac/Admin/guest/.lpoptions              0660  501  500  1079131620  23     1rwlsnjWK1+yjbb154i3fMeQXOc=
d ./private/etc/cups/ppd/                         0755  0    26
f ./private/etc/cups/ppd/_155.97.12.222.ppd       0644  0    26   1079131553  64619  kSLIxTgO8qcG79FIRcfDPBNRw4Y=
f ./private/etc/cups/ppd/_155.97.12.223.ppd       0644  0    26   1079131482  64619  kSLIxTgO8qcG79FIRcfDPBNRw4Y=
f ./private/etc/cups/printers.conf                0600  0    26   1079131553  493    qTZowwo9xvzvzpI7cgKhr9x84kY=
f ./private/etc/printcap                          0644  0    0    1079131553  282    SSFKeIDl8dNMvJJ3nKzMChFIlBs=
```



Negative Transcript

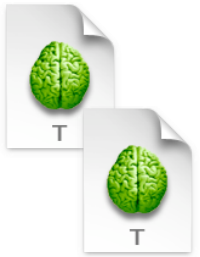
- Lists objects that **ARE NOT FULLY** managed
  - But they must exist managed client
  - Only some attributes check
    - For directories only UID & GID set
    - Directory contents are NOT read
    -
- Areas to used negative transcripts
  - Continually changing areas of file system
    - /dev
    - /Library/caches
  - Areas you want to retain for admin purposes
    - /private/var/log
  - End Users Needs
    - /Users
    - /private/etc/httpd/users
  - Software issues or logs





Negative Transcript

- Multiple Negatives
  - You can use multiple negatives per file system description
    - Baseline Negative
      - Defines OS and administration policy
    - Software Specific
      - Settings, logs, etc.



- Negatives can be placeholder or defaults
  - Placeholder is empty file (0 byte)
    - Useful when object must exist, but don't need default
    - Many of the OS's negative objects are setup this way



  - Defaults Settings
    - Useful when a file must exist
    - But need default settings for initial distribution
    - Applications like Retrospect Client 6 "retrospect.state"

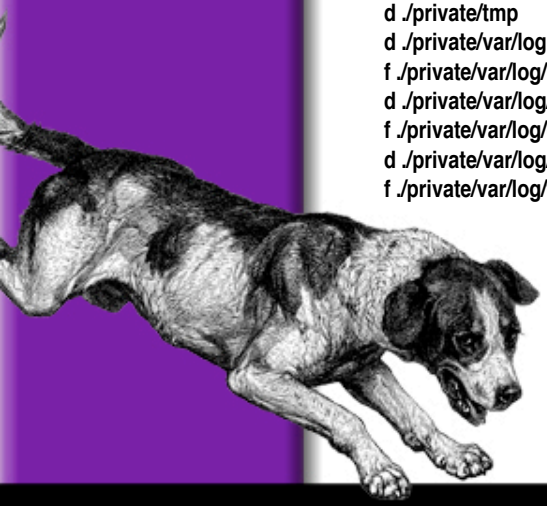




Negative Transcript

- Example of Mac OS X 10.3.5 negative

```
a ./.hotfiles.btree                                                0600  0  80  1071528930  94 e25XUAMeUwitvZY5DnLvqOCiGy8=
d ./.vol                                                           0444  0  0
d ./Library/Caches                                                 1777  0  80
d ./Library/Caches/com.apple.ATS                                   0777  0  80
d ./Library/Logs                                                   0775  0  80
d ./Library/Logs/Console                                           0775  0  80
d ./Library/Logs/DirectoryService                                  0775  0  80
f ./Library/Logs/DirectoryService/DirectoryService.server.log      0644  0  80  1087917478  0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=
f ./Library/Preferences/SystemConfiguration/preferences.plist      0644  0  80  1071620890  0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=
d ./Network                                                        0755  0  0
d ./System/Library/Caches                                          0755  0  0
f ./System/Library/Extensions.kextcache                            0644  0  0   1071528930  0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=
f ./System/Library/Extensions.mkext                                0644  0  0   1071528930  0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=
d ./Volumes                                                        1777  0  80
d ./automount                                                      0755  0  80
d ./dev                                                            0555  0  0
f ./mach.sym                                                       0444  0  80  1071528930  0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=
d ./private/automount                                              0755  0  80
d ./private/tmp                                                    1777  0  0
d ./private/var/log                                                0755  0  0
f ./private/var/log/CDIS.custom                                    0644  0  0   1087915051  17 cdpbznGU4SZ2P2kXrnSUGn+3nKg=
d ./private/var/log/cups                                           0755  0  0
f ./private/var/log/cups/error_log                                 0644  0  26  1087917468  0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=
d ./private/var/log/fax                                            0755  0  0
f ./private/var/log/ftp.log                                        0640  0  80  1063399303  0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=
```



Negative Transcript



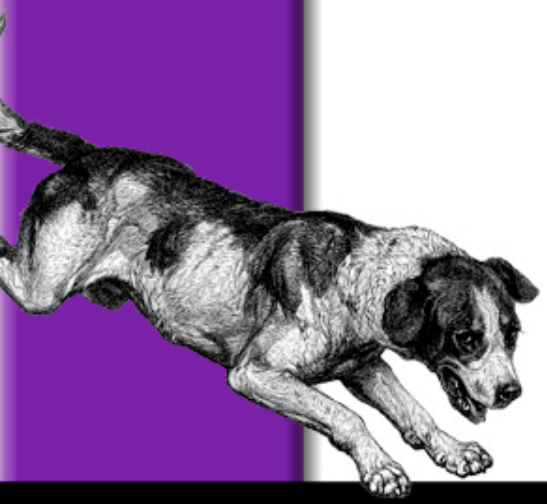
Example of Retrospect Client 6.x negative

f ./Library/Preferences/retroclient.state 0644 0 80 1097269240 2136 32ZFmSaVd/MyOLCZ2ZXt5ZZMvK0=



Example of Timbuktu Pro 7.x negative

f ./Library/Preferences/Netopia/Timbuktu\bPro\bLog 0640 0 80 1015476977 0 2jmj7l5rSw0yVb/vlWAYkK/YBwk=



Apply-able Transcript

- Differences on Client
  - Differences between the managed & local file system
- Modifications that need to applied
  - These are the modifications that must be applied to the client for it to match the managed file system.
- Restore Client
  - Used to restore client to managed state



Apply-able Transcript

- Apply-able Transcript Designators
  - **+** Missing files on client, download form server
  - **-** Remove files on client
  - Lines listed with no +/- indicate file system objects
    - Need to be created (like directories)
    - Or modified (like permissions & attributes)

```
+ f ./private/var/db/locate.database          0444  -2  0  1077579181  4750407  UdU+YlkRV1hP5T8yra8c1gt2PQQ=
- f ./private/var/db/netinfo/local.nidb/Clean 0644  0   0  1097995071  4        51kJPHsyMa+13GHfDR7cPRoX4W8=
f   ./private/var/db/netinfo/local.nidb/Config 0644  0   0  1084912712  4        R54E89EtEStcBMnuZ+Sx5ulB6k4=
```



Create-able Transcript

- Additions on Client
  - Is a transcript containing the additions on the client NOT on the managed file system.
- Additions Created
  - These are the additions that must be created on the deployed file system to match the client.
- Used to create baseload & overloads
- Add software, updates or modifications to file system



Other notable terms & concepts

- Loadset
  - Transcript & Associated Files
- Baseload
  - Primary large loadset
- Overload
  - Additional smaller loadsets
- Command File
  - Describes the clients managed file system
- Config File
  - Lists managed clients & file system



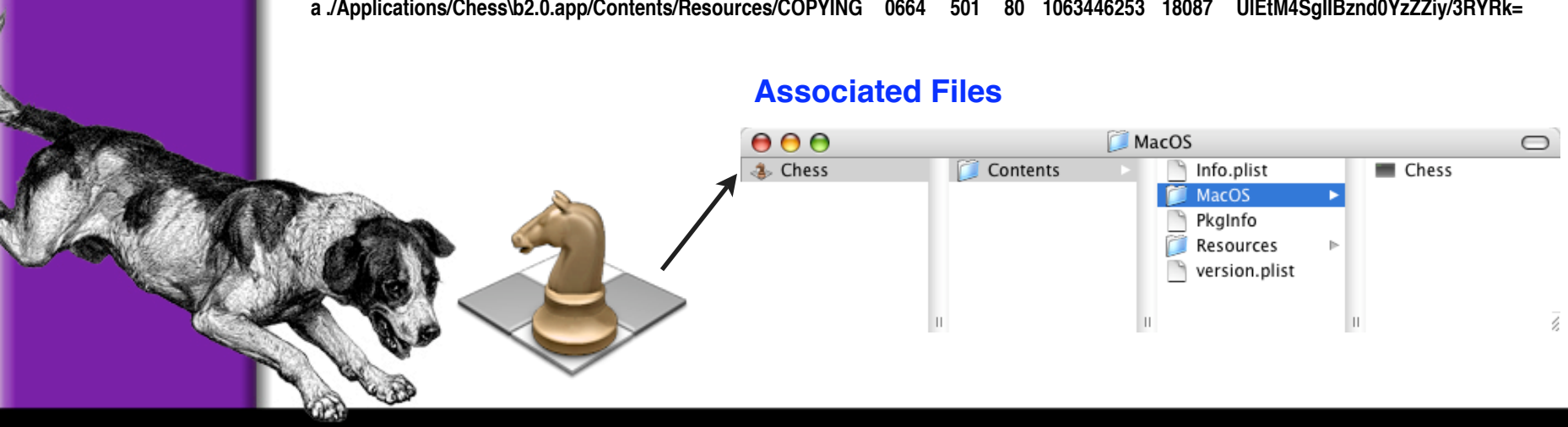
Loadset

- The transcript & associated file system objects

Transcript

```
d ./Applications/Chess\b2.0.app                                  0775  501  80
d ./Applications/Chess\b2.0.app/Contents                         0775  501  80
a ./Applications/Chess\b2.0.app/Contents/Info.plist              0664  501  80  1063446253  1829    a8NJHudyyx6pfT9oHNVaykNSiNc=
d ./Applications/Chess\b2.0.app/Contents/MacOS                   0775  501  80
a ./Applications/Chess\b2.0.app/Contents/MacOS/Chess             0775  501  80  1096299220  220822  g2Zf67mmUAZR1a1s4ykg+my9BHQ=
a ./Applications/Chess\b2.0.app/Contents/PkgInfo                 0664  501  80  1063446253  102     9Kw3S/TT9JC9BoRJ5HbGF9Id2vw=
d ./Applications/Chess\b2.0.app/Contents/Resources               0775  501  80
a ./Applications/Chess\b2.0.app/Contents/Resources/1.rgb         0664  501  80  1063446253  2740    K2+Ub1bA9iArGBkl7tjyGNs9gvA=
a ./Applications/Chess\b2.0.app/Contents/Resources/2.rgb         0664  501  80  1063446253  3037    ZocJtCPnu0NH0BSw92MaqlBFeSA=
a ./Applications/Chess\b2.0.app/Contents/Resources/3.rgb         0664  501  80  1063446253  3040    CGZPshzPPOoQyfbNZqnWUTal7W4=
a ./Applications/Chess\b2.0.app/Contents/Resources/4.rgb         0664  501  80  1063446253  2971    KlkvasubxAFrT4RAsnHN7TR2bQQ=
a ./Applications/Chess\b2.0.app/Contents/Resources/5.rgb         0664  501  80  1063446253  3082    w/C/hB03jMAVVG8KYXR3RzyNcns=
a ./Applications/Chess\b2.0.app/Contents/Resources/6.rgb         0664  501  80  1063446253  3157    iklyWKAmpJHfbiF5uWFY3djkfB4=
a ./Applications/Chess\b2.0.app/Contents/Resources/7.rgb         0664  501  80  1063446253  2959    6/vX4bgrKktSHlvSE4eloMc1lCk=
a ./Applications/Chess\b2.0.app/Contents/Resources/8.rgb         0664  501  80  1063446253  3052    zmRPzkAgoQHnZmCxXhkASFvKNVl=
a ./Applications/Chess\b2.0.app/Contents/Resources/COPYING       0664  501  80  1063446253  18087   UlEtM4SgllBznd0YzZZiy/3RYRk=
```

Associated Files



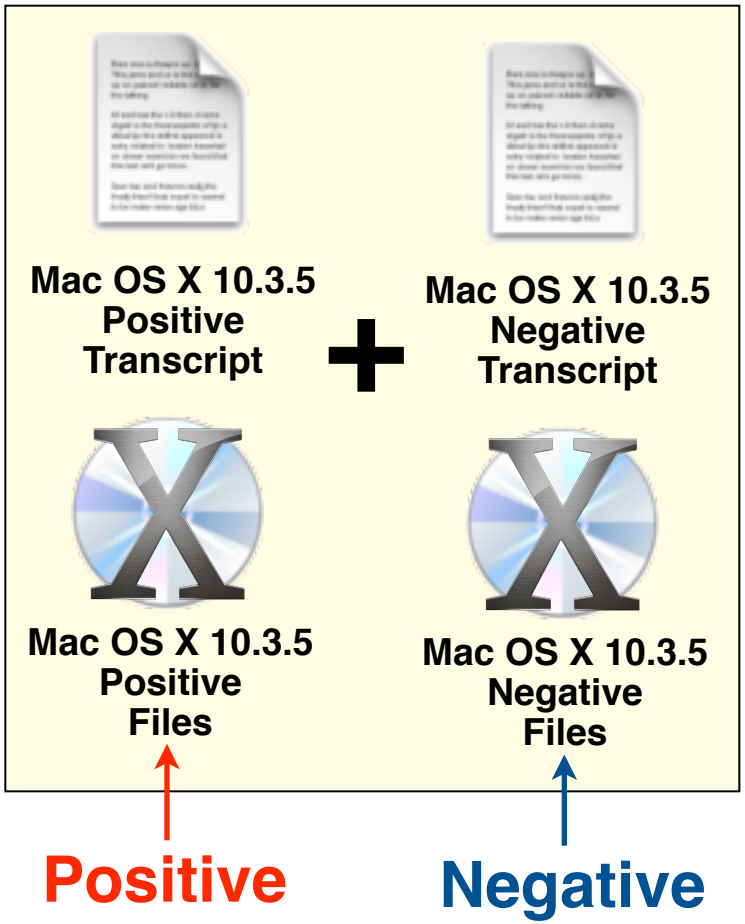
Baseload

- The primary file system loadset
- Usually a large transcript & associated files.
  - A baseload could be:
    - Operating System
    - Operating System and global applications
    - Or the entire client file system deployment
      - Including operating system
      - Applications
      - Admin Modifications
        (users & groups, directory setup, scripts, etc.)



Baseload - Example

- Baseload that only contains OS





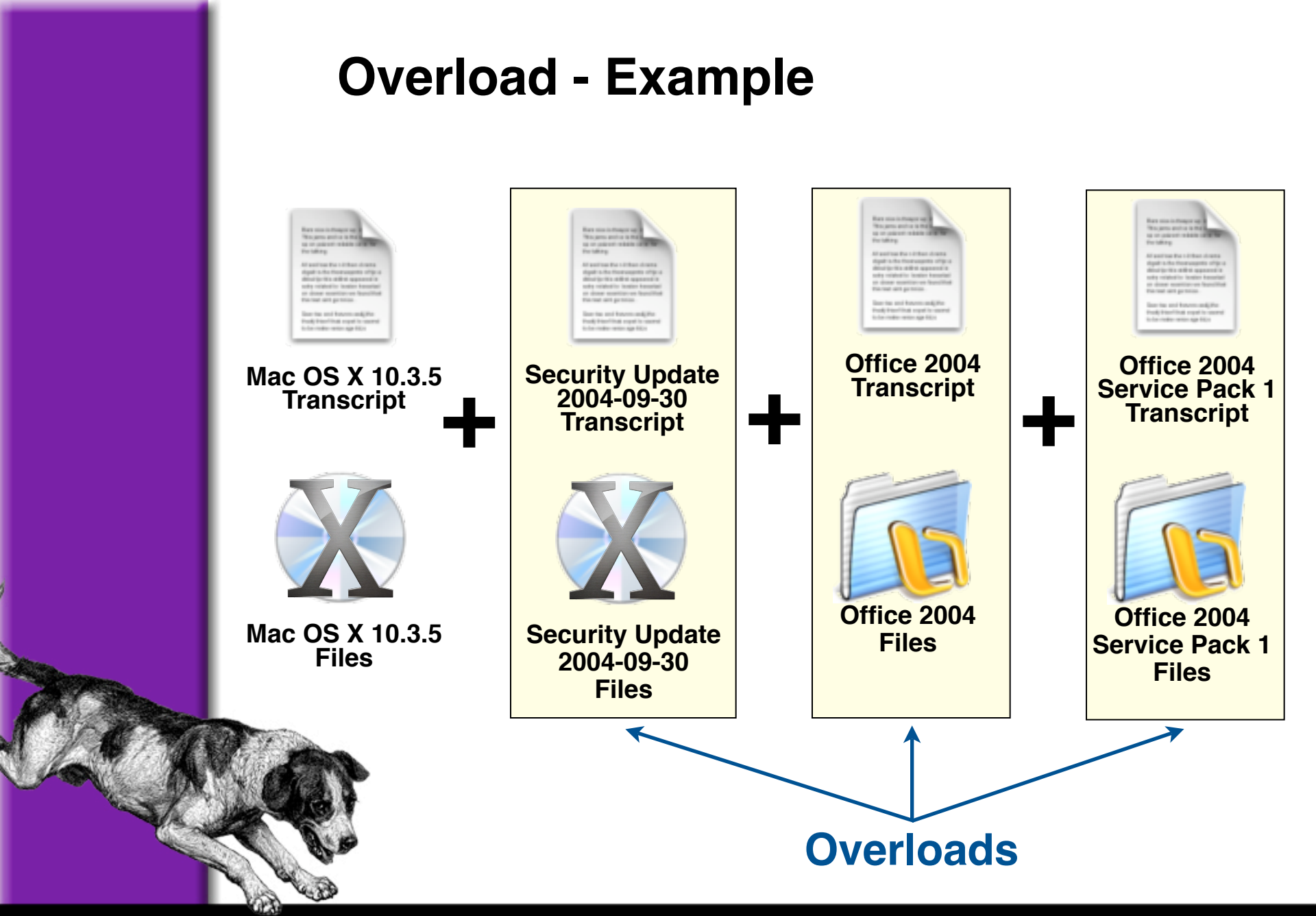
Overload

- A smaller additional loadset used with baseload

- Examples:
  - Software (Office 2004 & Service Pack 1)
  - Software/OS (Security Update 2004-09-30)
  - Settings (Lab Printer configuration)
  - Individual Objects (Custom Dock - com.apple.dock.plist)



Overload - Example



Command File - General

- Defines the client(s) managed file system
- A text file listing the baseload & additional overloads
- Command files are named in the format:

  <command_file_name>.K

  – Can't use spaces in names
- On sever command files can have various names:
  – kiosk.K
  – lab.K
  – staff.K
  – john_doe.K

- But, by default, on client it will be renamed to:
  – command.K



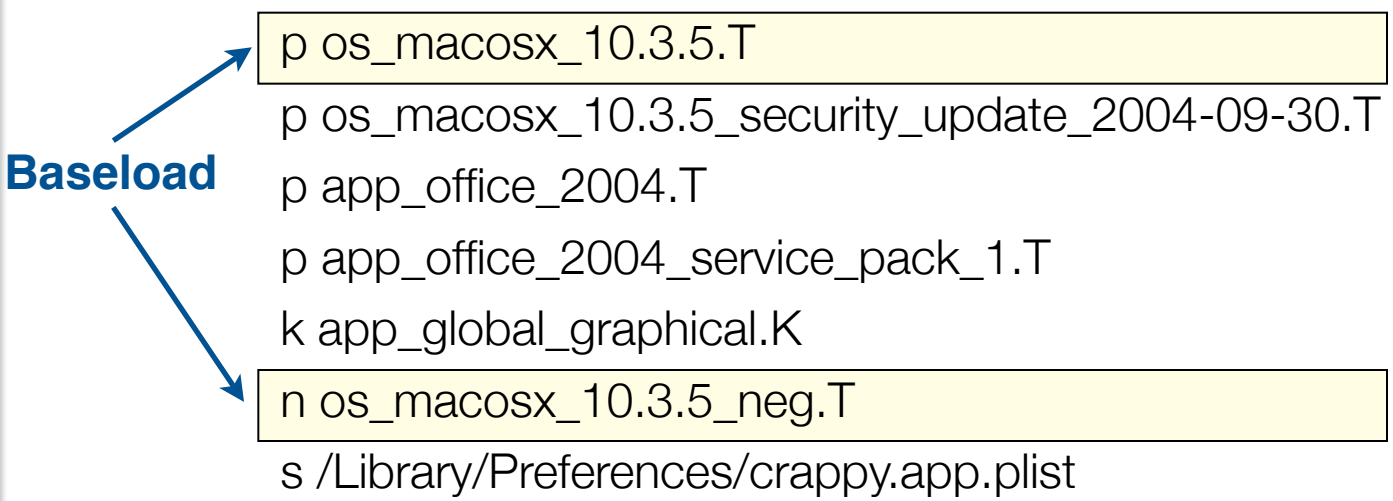
Command File - Designators

- Uses designators to denote transcript type
- This type is a single letter:
  - **p** positive transcript
  - **n** negative transcript
  - **s** special transcript
- Special transcripts appear at the absolute end
- A new type that is available in beta is:
  - **k** command file
  - This allows to group & organize transcripts
  - You can have on command file group transcripts
    - List it only in that command file
    - And list that command file in other command files



Command File - Baseload

- Example of command file, called lab.K:





Command File - Overloads

- Example of command file, called lab.K:

Overloads

```
p os_macosx_10.3.5.T
p os_macosx_10.3.5_security_update_2004-09-30.T
p app_office_2004.T
p app_office_2004_service_pack_1.T
k app_global_graphical.K
n os_macosx_10.3.5_neg.T
s /Library/Preferences/crappy.app.plist
```



Command File - k

- Example of command file, called lab.K:

Command File

```
p os_macosx_10.3.5.T
p os_macosx_10.3.5_security_update_2004-09-30.T
p app_office_2004.T
p app_office_2004_service_pack_1.T
k app_global_graphical.K
n os_macosx_10.3.5_neg.T
s /Library/Preferences/crappy.app.plist
```



Command File - k Example

- Example of app_global_graphical.K:

```
p app_after_effects_6.0.T
p app_freehand_10.0.T
p app_freehand_mx_11.0.1.T
p app_graphicconverter_5.2.1.T
p app_illustrator_10.0.3.T
p app_illustrator_cs_11.0.T
p app_painter_7.1.T
p app_photoshop_7.0.1.T
p app_photoshop_cs_8.0x119.T
p app_photoshop_elements_2.0.T
p app_quarkxpress_passport_6.1.T
```



Command File - Special

- Example of command file, called lab.K:

```
p os_macosx_10.3.5.T
p os_macosx_10.3.5_security_update_2004-09-30.T
p app_office_2004.T
p app_office_2004_service_pack_1.T
k app_global_graphical.T
n os_macosx_10.3.5_neg.T
s /Library/Preferences/crappy.app.plist
```

Special



Command File - Order of Precedence

- Use order of precedence
  - First transcript listed is lowest precedence
  - Last transcript listed is highest precedence
  - Example of command file, called lab.K:

Lowest

p os_macosx_10.3.5.T
p os_macosx_10.3.5_security_update_2004-09-30.T
p app_office_2004.T
p app_office_2004_service_pack_1.T
k app_global_graphical.K
n os_macosx_10.3.5_neg.T

Highest

s /Library/Preferences/crappy.app.plist

  - This order defines which items are used when there are duplicates or conflicts and other transcripts.



Command File - Office 2004

- Order of precedence
  - For example we have two transcripts
    - First, is for the original Office 2004 install
    - Second, is for the Office 2004 update

Original → p app_office_2004.T
Update → p app_office_2004_service_pack_1.T

  - The update will include items listed in the original install
  - But, since the update is listed below the original
  - It takes precedence over the original
  - So, the duplicated items will be used from the update **NOT** the original.



Config File

- A text file on the radmind server
- Lists known clients that can connect to sever
- Clients are identified by:
  - IP Address
  - Hostname
  - SSL Certificate
  - Rendezvous
- Hostnames & certificate names ARE case sensitive

- Lines that are blank or begin with “#” are ignored

- Identifying a group or range of clients
  - A number range can be given by "<min-max>"
  - * is a wildcard & will match any string



Config File

- For example...

richard.scl.utah.edu	tech_richard_lab.K
169.192.1.253	staff_john.K
james.scl.utah.edu	tech_james_lab.K
ken.scl.utah.edu	staff_ken.K
*.lab.utah.edu	lab.K
169.192.1.<1-50>	kiosk.K

  - Examples of using hostnames, individual IP's addresses, IP ranges, and wildcards to identify clients.



Radmind Tools

Overview of radmind command-line tools



fsdiff

- File System Differences
  - The primary radmind tool
  - Compares the file system to a set of transcripts described by a command file
  - Creates the differences in the form of a transcript
  - Never talks on the network
  - Detects file system modifications (tripwire)



fsdiff - Example

Create-able transcript named loadset.T

Terminal — tcsh (ttyp1)

```
[client] fsdiff -C -c sha1 -o /var/radmind/client/loadset.T /
```

Apply-able transcript named differences.T

Terminal — tcsh (ttyp1)

```
[client] fsdiff -A -c sha1  / > /tmp/differences.T
```



fsdiff - Example

Create transcript line of single item

Terminal — tcsh (ttyp1)

```
[client] fsdiff -c sha1 -1 /path/to/item
```

Create transcript lines of contents of directory

Terminal — tcsh (ttyp1)

```
[client] fsdiff -K /dev/null -c sha1 /path/to/directory
```



lcreate

- loadset create
  - Uploads a transcript and associated files to the server
  - Stores Mac OS dual forked files as AppleSingle encoded files on the server
  - This encoding preserves the data & resource fork
    - Allows files to be stored on single forked file systems



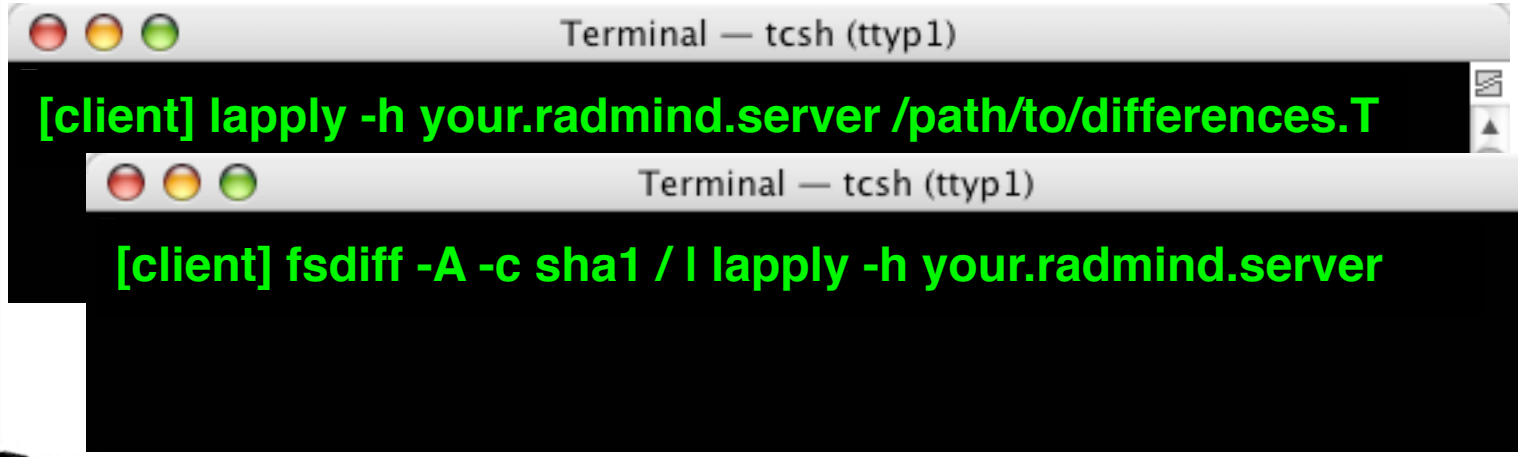
Uploads transcript loadset.T with associated files



lapply

- Loadset Apply
  - Modifies local file system as described by an apply-able transcript
  - Retrieves updated files as required from the server
  - Updates client to managed state

lapply using differences file

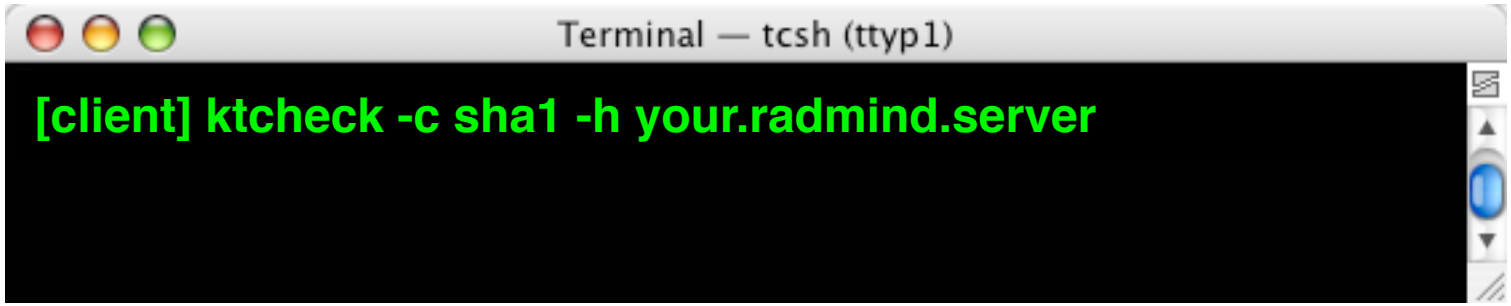


lapply using pipe



ktcheck

- Command & Transcript Check
  - Compares local command file and associated transcripts with those stored on the server
  - Retrieves updated command files and transcripts from the server

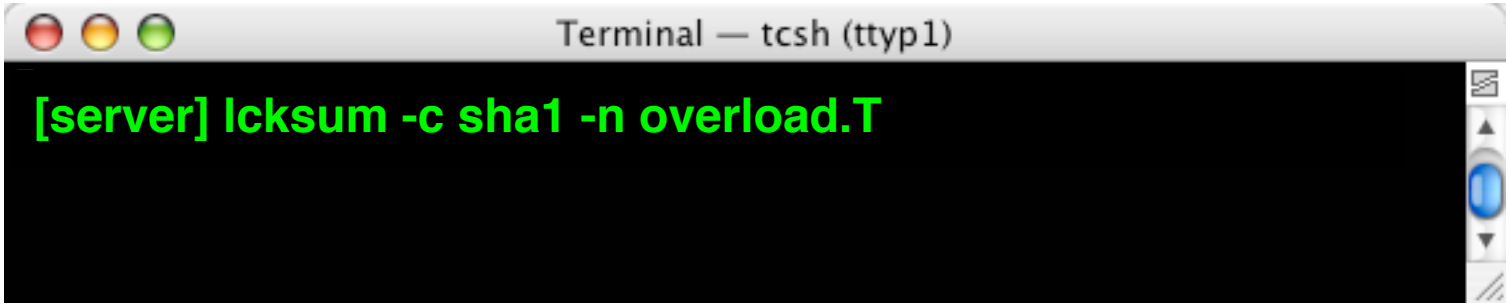


ktcheck example

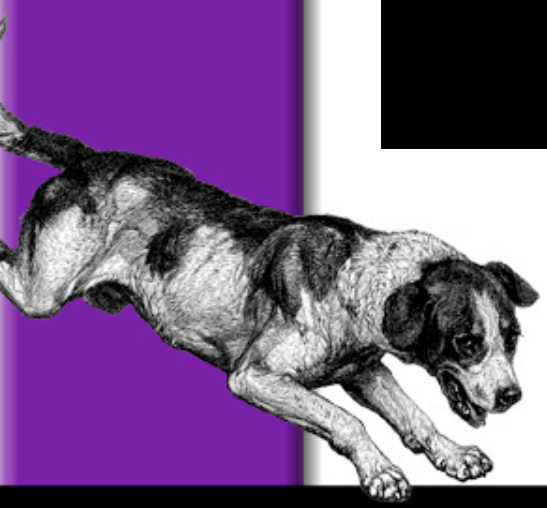


lcksum

- Loadset Checksum
  - Verifies & updates the checksum & size of files associated with a transcript on the server
  - With -n option verifies but does not update transcript
  - Also verifies that transcript is sorted in depth first order



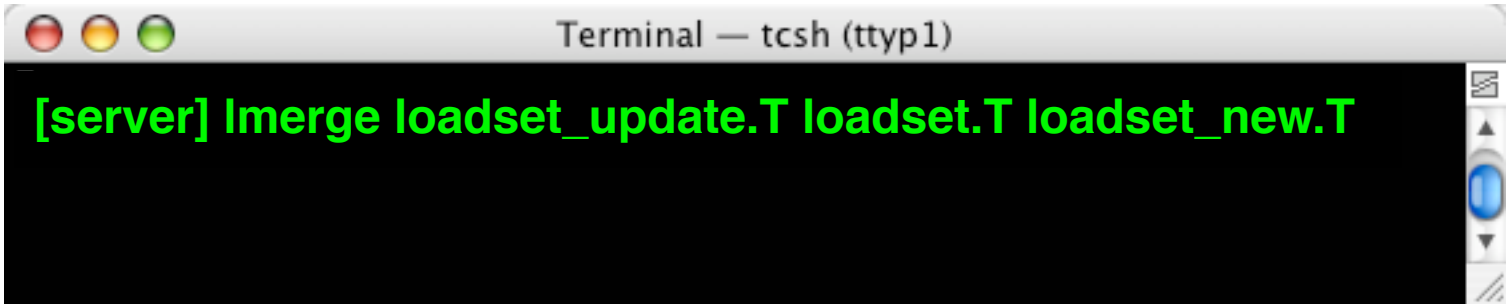
lcksum example



Imerge

- Loadset Merge
  - Combines two or more transcripts and associated files into one loadset

Imerge example







loadset.T
Mac OS X 10.3.5

loadset_update.T
Security Update 2004-09-30

loadset_new.T
Mac OS X 10.3.5 w/update

twhich

- Which Transcript
  - Searches each transcript listed in a command file for a specified object.
  - Using option -a it lists all transcripts that contain object
  - This tool is useful to debug conflicts between transcript lines in a command file



twhich example



Brief Tutorial

Initial Setup, Negative, Baseload, Update

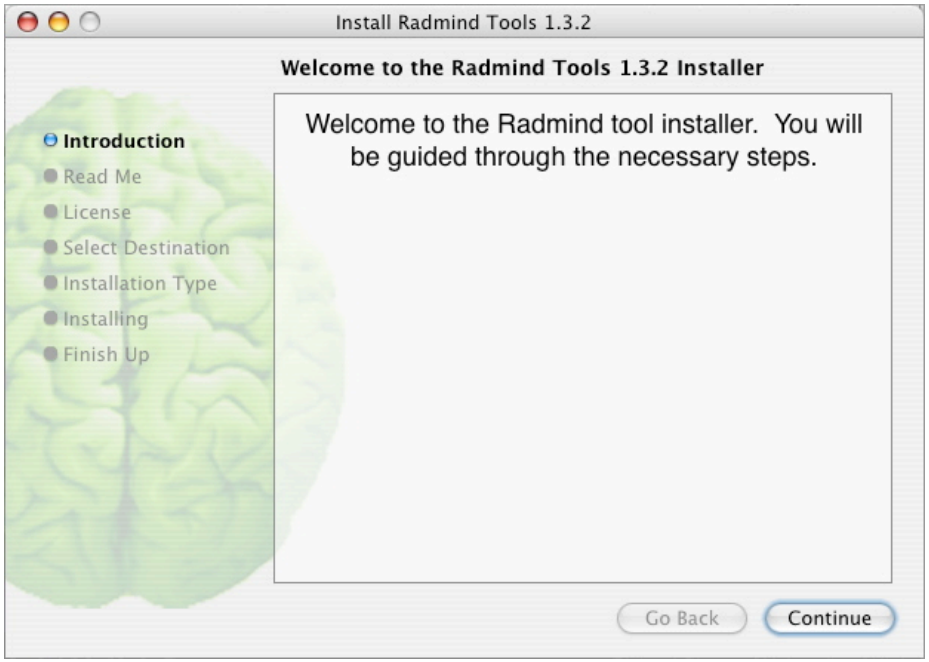


Initial Install



Client

- Setup your client to how you want the file system
- Download the latest Mac OS X Installer
- Run installer...





Install Locations



Client

- Radmind tools are installed at /usr/local/bin
- man pages are installed at /usr/local/man
- Creates directories

  /var/radmind/

  /var/radmind/client

  /var/radmind/cert

- You can create missing directories:

  [client] mkdir /var/radmind/client

  [client] mkdir /var/radmind/cert



Base Negative Transcript



Client

- Next, setup a baseline negative transcript
- Example negative transcripts available
  - University of Utah - Radmind site:

    www.macos.utah.edu/Documentation/radmind/resources.html#negative_examples
  - Radmind Site

    www.radmind.org/macosx/
- Pick a example negative & edit it with text editor
  - Comment or uncomment lines in examples
  - Use fsdiff -1 /path/to/item to add additional items
  - Change directory to /var/radmind/client
  - Save transcript as base_negative.T



Negative Transcript Example



Client

RSUG-MacOSX_10.3_lab_negative.T

4493160 bytes 52 lines

Line #	Type	Path
1		# A template negative transcript for managing machines running
2		# Mac OS X 10.3 in a lab setting.
3		#
4		# Do not use this file without first examining it thoroughly.
5		#
6		# Lines starting with '#' and blank lines are ignored.
7		# Stores trash on the root volume
8	d	/.Trashes
9		# Apparently associated with journaling
10	a	/.hotfiles.btree
11		# Used by the system and NFS
12	d	/.vol
13		# Caches and other directories containing volatile items
14	d	/Library/Caches
15	d	/Library/ColorSync/Profiles/Displays
16	d	/Library/Logs
17	d	/Library/Preferences/SystemConfiguration
18	d	/Network
19	d	/System/Library/Caches
20		# Kernel extensions cache files. Created on boot if missing or
21		# older than /System/Library/Extensions
22	f	/System/Library/Extensions.kextcache
23	f	/System/Library/Extensions.mkext
24		# User space. Let the user management scripts manage it for you.
25	d	/Users
26		# mount point for other drives



Client Initial Command File



Client

- Next, we need to create/edit a command file to use for the initial baseload setup
  - On the client,

    cd /var/radmind/client
  - Then create an empty command file

    touch command.K
  - Or use your favorite text editor (BBEdit)
- Next, we need to add the negative transcript

  n base_negative.T
- And save the command.K file



Create Positive Baseload



Client

- Next, create your positive baseload for the file system

```
fsdiff -C -c sha1 –o /var/radmind/client/base_positive.T /
```

- fsdiff uses the command file the we previously created
- Depending on hardware & the size of your file system
  - Note, fsdiff could take a minute to 40 minutes
- Once it is done it will return you to the prompt



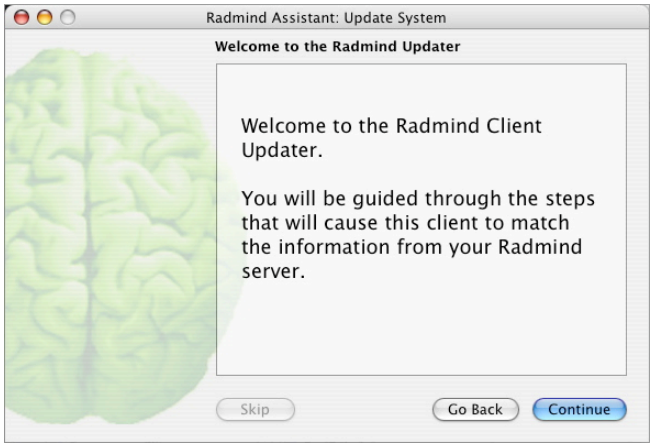
Setup Server



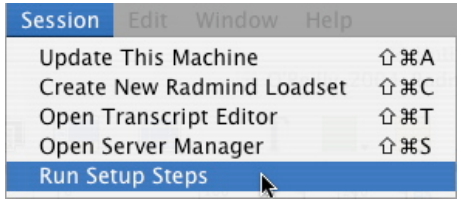
Server

- Run the radmind installer on your server
- Download and launch the Radmind Assistant





- Select “Run Setup Steps” from the “Session” menu





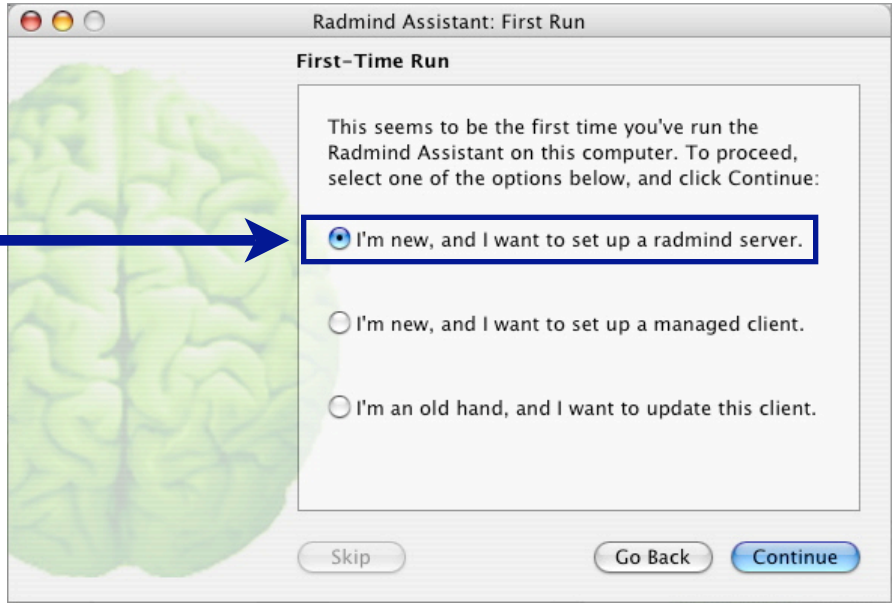
Setup Server



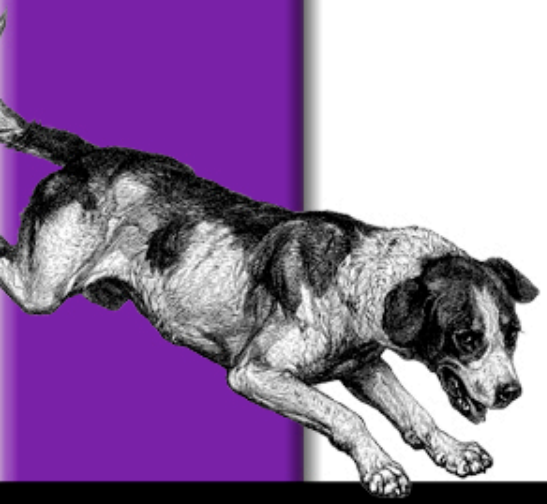
Server

- Next, the following window will display...

Select this option



- Select the option:
  - I'm new, and I want to setup a radmind server.
- Click the"Continue" button



Setup Server

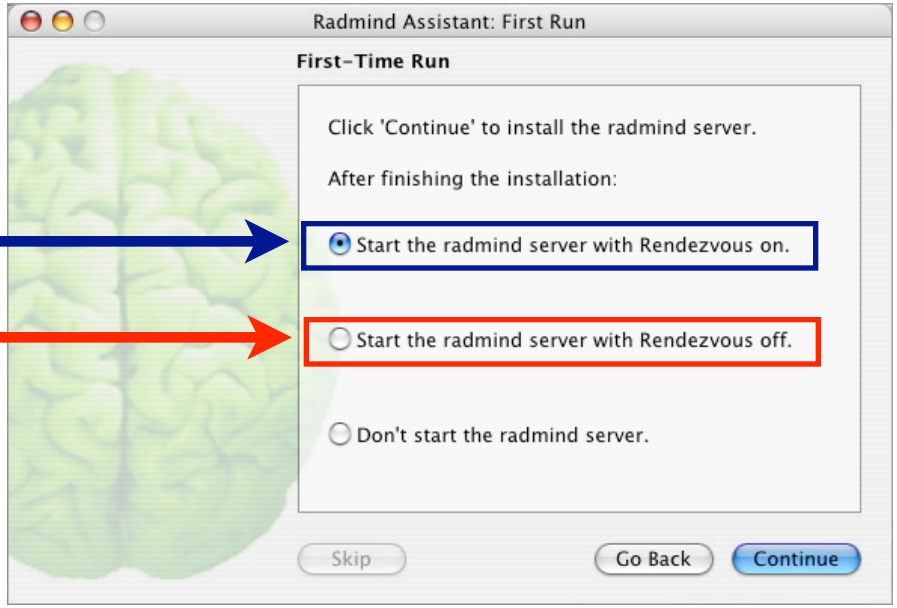


Server

- Next, the following window will display...

Rendezvous On

Rendezvous Off



- Two options start server with rendezvous on or off.
- Click the“Continue” button



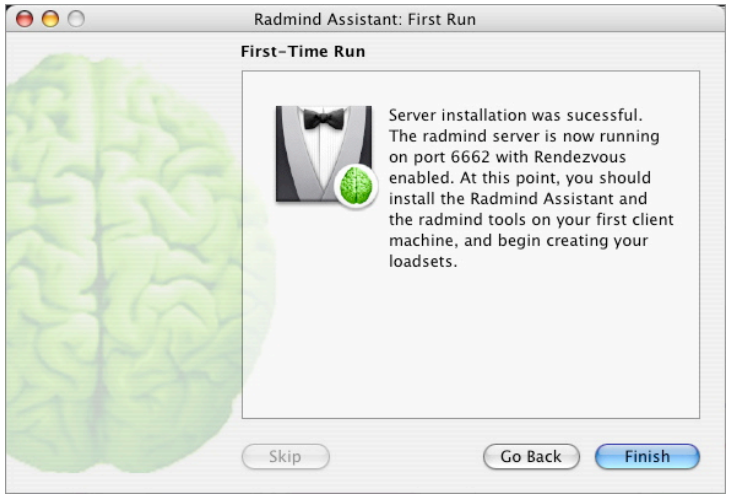
Setup Server



Server

- You will be prompted to authenticate
  - Enter admin username/password
- If there weren't any errors
- You will get the following dialog...





- Then quit Radmind Assistant



Setup Server



Server

- What just happened?
- Radmind daemon was started...

```
/usr/local/sbin/radmind -u 077
```

- StartupItems directory & items were created

Directory → /Library/StartupItems/RadmindServer

Script → /Library/StartupItems/RadmindServer/RadmindServer

Startup Parameters → /Library/StartupItems/RadmindServer/StartupParameters.plist

- Radmind daemon run on TCP port 6662



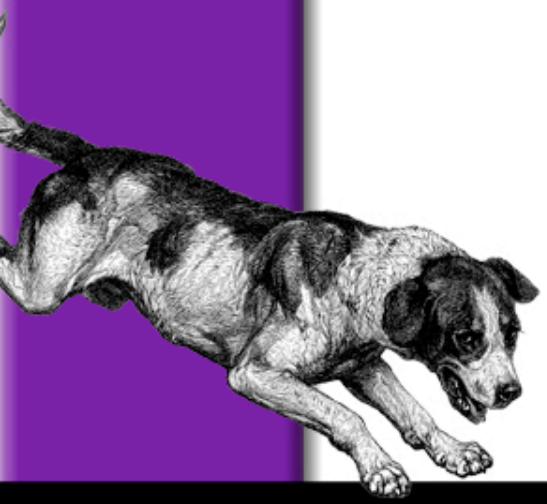
Setup Server



Server

- What just happened?
- The radmind deamon created the following directories:

/var/radmind/command	**Client Command Files**
/var/radmind/file	**Loadset Associated Files**
/var/radmind/special	**Host Specific**
/var/radmind/transcript	**Loadset Transcripts**
/var/radmind/tmp	**Temporary Directory**
/var/radmind/tmp/transcript	**Temp Transcripts**
/var/radmind/tmp/file	**Temp Associated Files**



Setup Server



Server - Script

- Here is the StartupItem script...

```
#!/bin/sh

. /etc/rc.common

##
# Start up radmind server
##

if [ -f /var/radmind/config ]; then
    CheckForNetwork

    if [ "${NETWORKUP}" = "-NO-" ]; then exit; fi

    ConsoleMessage "Starting Radmind Server"

    /usr/local/sbin/radmind -u 077

fi
```



Setup Server



Server - Startup Parameters

- Here is the StartupParameters.plist...

```
{
  Description     = "Radmind Server";
  Provides        = ("Radmind Server");
  Requires        = ("Resolver");
  OrderPreference = "None";
  Messages =
  {
    start = "Starting radmind server";
    stop  = "Stopping radmind server";
    restart = "Restarting radmind server";
  };
}
```

- Change the following line...

```
Provides        = ("Radmind Server");
```

- To...

```
Provides        = ("Radmind Server", "Disks"));
```



Setup Server



Server - Client Command & Config Files

- Next, we need to create a command file
  - On the server, change directory to...

    cd /var/radmind/command
  - Then create an empty command file

    touch base.K
  - Or use your favorite text editor (BBEdit)
- Next, we need to identify the client
  - On the server, open the config file

    /var/radmind/config
  - Add the IP Address or hostname of client

    169.192.1.253 base.K
  - And save config file



Upload Base Negative & Positive



Client

- Next, go back to the client
- And upload the base negative & positive
- First, start with the negative base loadset

```
lcreate –N –h your.radmind.server /var/radmind/client/base_negative.T
```

- The -N option stored files as zero length
- Next, start uploading the positive base loadset

```
lcreate –h your.radmind.server /var/radmind/client/base_positive.T
```

- Note, depending on the size, Mac model, network, etc.
  - The could take 2-6 hrs, so, go to lunch or see a movie



Verify Loadset



Server

- The loadset is uploaded to a temporary location
- The transcript will be in directory:
  /var/radmind/tmp/transcript
- And the associated files willl be in directory:
  /var/radmind/tmp/file
- Change to the temporary transcript directory
  cd /var/radmind/tmp/transcript
- To verify loadset enter the following command:
  lcksum -c sha1 base_negative.T
  lcksum -c sha1 base_positive.T



Move Loadset into Production



Server

- After the loadset has been verified
- The next step is to move it into production

- To move transcript:

```
mv /var/radmind/tmp/transcript/base_negative.T /var/radmind/transcript/
mv /var/radmind/tmp/transcript/base_positive.T /var/radmind/transcript/
```

- And move the associated files:

```
mv /var/radmind/tmp/file/base_negative.T /var/radmind/file/
mv /var/radmind/tmp/file/base_postivie.T /var/radmind/file/
```



Edit Command File



Server - Add transcripts to command file

- Next, you need to edit command file for client
- Open the command file base.K

  /var/radmind/command/base.K

- Add add the following lines for the transcripts

  p base_positive.T
  n base_negative.T

- You can use your favorite command line text editor
- Or GUI text editor, like BBEdit





Edit Command File



Server - Test Loadset

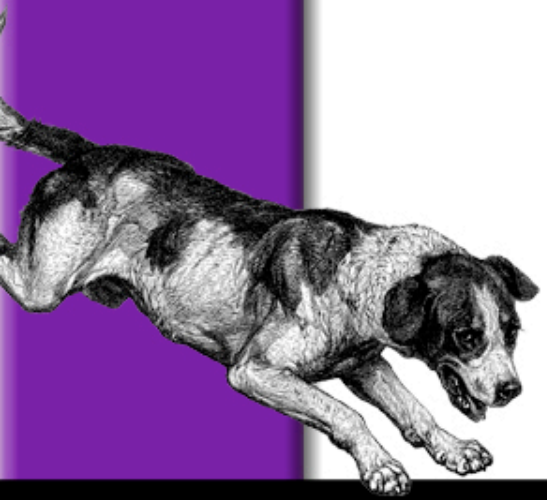
- Next, we want to test the loadset on another client
- Edit the config file and add the IP or hostname

  /var/radmind/config

- For example...

  169.192.1.253 base.K ← **Original Client**
  169.192.1.254 base.K ← **Test Client**

- Add test client and assign command file base.K
- Save config file



Update Client



Client - Testing the baseload

- Next, we want to test the base loadset on another Mac
- Install the radmind tools
- Update client command & associated transcripts

```
ktcheck -c sha1 -h your.radmind.server
```

- Create apply-able transcript & apply it to client

```
fsdiff -A -c sha1 / | lapply -h your.radmind.server
```

- Once it is done, reboot and test client



Related Sessions

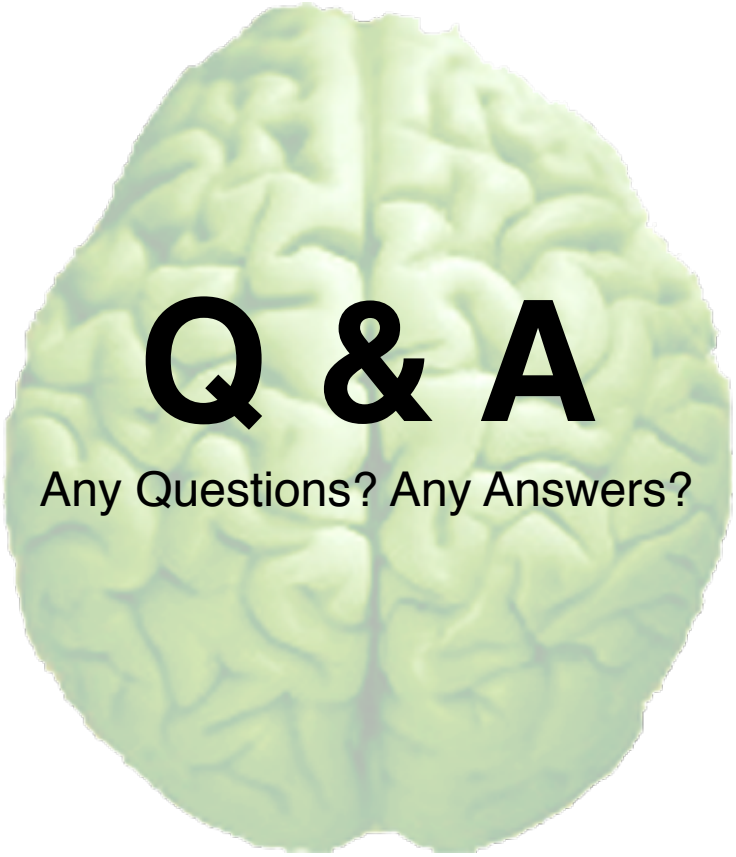


Related Sessions

- **Automating Computer Management**
  Wed, 11:25 AM - 12:20 PM

  Will covering a suite of scripts, tools, graphics used to managed labs at the University of Utah.





Q & A

Any Questions? Any Answers?



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